

Radar_Sensor_Preprocessing Unit Test Specification

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Chapter 1

Covered Requirements

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NA.

Member [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_positive](#) ()
NA.

Member [compute_detection_properties_Compute_Detection_VCS_Coordinates](#) ()
NA.

Member [compute_detection_properties_Test_Updated_detection_Time_with_data](#) ()
NA

Member [f360_check_det_az_conf_and_elevation_FunctionToTest_Descriptive_Tag](#) ()
Put in requirement tag if any otherwise set to NA.

Member [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation](#) ()
FTCP-8842

Member [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1](#) ()
FTCP-8842

Member [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2](#) ()
FTCP-8842

Member [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3](#) ()
FTCP-8842

Member [f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off](#) ()
FTCP-8842

Member [f360_clear_dets_vcs_long_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#) ()
NA

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()
FTCP-8190

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()
FTCP-8190

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification](#) ()
FTCP-8190

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classifi](#)
 ()
 FTCP-8190

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classifi](#)
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 FTCP-8190 NA.

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification](#)
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 FTCP-8190

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 FTCP-8190

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 FTCP-8190

Member [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp](#)
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Member `f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_comp`
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 FTCP-8402

Member `f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp`
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 FTCP-8402

Member `f360_filter_detections_with_bad_azimuth_confidence_FunctionToTest_Descriptive_Tag` ()
 Put in requirement tag if any otherwise set to NA.

Member `f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_General_Case`
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Member `f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Position_Update` ()
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NA

Member [f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_Non_Zero_Slope \(\)](#)
NA

Member [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_No_Slope \(\)](#)
NA

Member [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_Non_Zero_Slope \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_1 \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_2 \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Residual_Value_Slightly_Smaller_Than_Tuning_Constant \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Slightly_Larger_Than_Tuning_Constant \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Equal_To_Tuning_Constant \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Greater_Than_Tuning_Constant \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Slightly_Greater_Than_Tuning_Constant \(\)](#)
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator](#)
()
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero](#)
()
NA

Member [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Slightly_Smaller_Than_Min_Denominator](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Correct_Result_For_Valid_Input](#) ()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Residual_Equal_To_Zero_And_Non_Zero_Tuning_Constant](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Equal_To_Positive_Residual](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Absolute_Value_Of_Residual](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Positive_Residual](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Absolute_Value_Of_Residual](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Positive_Residual](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Both_Input_Values_Equal_To_Zero](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Negative_Tuning_Constant](#) ()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Greater_Than_Min_Denominator](#)
()
NA

Member [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Smaller_Than_Min_Denominator](#)
()
NA

Member [f360_math_func_F360_Linear_Equation](#) ()
NA

Member [f360_math_func_F360_Linear_Equation_small_den](#) ()
NA

Member [f360_math_func_F360_Linear_Equation_With_Saturation_small_den \(\)](#)

NA

Member [f360_math_func_F360_Matmul_MxN_NxP \(\)](#)

NA

Member [f360_math_func_F360_Matmul_MxN_NxP_Transpose \(\)](#)

NA

Member [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_180 \(\)](#)

NA

Member [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_90 \(\)](#)

NA

Member [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90 \(\)](#)

NA

Member [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Zero \(\)](#)

NA

Member [f360_math_func_F360_Saturate \(\)](#)

NA

Member [f360_math_func_F360_Sort_asc \(\)](#)

NA

Member [f360_math_func_F360_Sort_dsc \(\)](#)

NA

Member [f360_math_func_F360_Sort_fail \(\)](#)

NA

Member [f360_math_func_F360_Sort_no_sort \(\)](#)

NA

Member [f360_math_func_F360_Sort_partial \(\)](#)

NA

Member [f360_math_func_F360_Sort_sorted_values_not_modified \(\)](#)

NA

Member [f360_math_func_F360_Sort_stability \(\)](#)

NA

Member [f360_math_func_F360_Sort_success \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Large_Slope \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Small_Slope \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Imaginary_Root \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Negative_A_Coefficient \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Positive_A_Coefficient \(\)](#)

NA

Member [f360_math_func_Find_X_Intersect_Between_Two_Lines_Intersection_Exists \(\)](#)
NA

Member [f360_math_func_Find_X_Intersect_Between_Two_Lines_Parallel_Lines \(\)](#)
NA

Member [f360_math_func_Get_Vector_As_Linear_Equation_Negative_Infinite_K_Zero_M \(\)](#)
NA

Member [f360_math_func_Get_Vector_As_Linear_Equation_Negative_K_Positive_M \(\)](#)
NA

Member [f360_math_func_Get_Vector_As_Linear_Equation_Positive_Infinite_K_Positive_M \(\)](#)
NA

Member [f360_math_func_Get_Vector_As_Linear_Equation_Positive_K_Zero_M \(\)](#)
NA

Member [f360_math_func_Test_F360_2d_Matrix_Determinant \(\)](#)
NA

Member [f360_math_func_Test_F360_Accumulate_Probabilities \(\)](#)
NA

Member [f360_math_func_Test_F360_Cross_Product_Of_Negative_Vectors \(\)](#)
NA

Member [f360_math_func_Test_F360_Cross_Product_Of_Parallel_Vectors \(\)](#)
NA

Member [f360_math_func_Test_F360_Cross_Product_Of_Vectors \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_negative_input \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_positive_input \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_squared_negative_input \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_squared_positive_input \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_squared_zero_input \(\)](#)
NA

Member [f360_math_func_test_f360_get_hypotenuse_zero_input \(\)](#)
NA

Member [f360_math_func_Test_F360_Linear_Equation_With_Saturation \(\)](#)
NA

Member [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order \(\)](#)
NA

Member [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas \(\)](#)
NA

Member [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas \(\)](#)
NA

Member [f360_math_func_Test_F360_Low_Pass_Filter_First_Order \(\)](#)
NA

Member [f360_math_func_Test_F360_matmul_2x2_2x1 \(\)](#)

NA

Member [f360_math_func_Test_F360_Mod_divisor_with_decimals \(\)](#)

NA

Member [f360_math_func_Test_F360_Mod_divisor_without_decimals \(\)](#)

NA

Member [f360_math_func_Test_F360_Multiply_2D_Matrix_With_Constant \(\)](#)

NA

Member [f360_math_func_Test_F360_Power_Scale_Prob \(\)](#)

NA

Member [f360_math_func_Test_F360_Safe_Logf \(\)](#)

NA

Member [f360_math_func_Test_F360_Translate_2D_Position \(\)](#)

NA

Member [f360_math_func_Test_Rotate_2D_Covariance_Matrix \(\)](#)

NA

Member [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5 \(\)](#)

NA

Member [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3 \(\)](#)

NA

Member [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle \(\)](#)

NA

Member [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3 \(\)](#)

NA

Member [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints \(\)](#)

NA

Member [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints \(\)](#)

NA

Member [f360_norm_heading_angle_FunctionToTest_Descriptive_Tag \(\)](#)

Put in requirement tag if any otherwise set to NA.

Member [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty \(\)](#)

NA

Member [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty_2 \(\)](#)

NA

Member [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_backward_speed \(\)](#)

NA

Member [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_Forward_speed \(\)](#)

NA

Member [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Inside_Range \(\)](#)

NA

Member [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_On_Boundary_Of_Range \(\)](#)

NA

Member [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Out_Of_Range](#)
 ()
 NA

Member [f360_sensor_capability_Get_Uncertainty_Of_Detection_Position](#) ()
 NA

Member [f360_sensor_capability_Sensor_Capability_Detections_Test_Sensor_Capability_Detections](#) ()
 NA

Member [f360_sensor_capability_Sensor_Capability_Host_Test_Sensor_Capability_Host](#) ()
 NA

Member [f360_sensor_capability_Test_Compute_Detection_Cross_Covariances](#) ()
 NA

Member [f360_sensor_capability_Test_Compute_Raw_Host_Speed_Uncertainty](#) ()
 NA

Member [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point](#)
 ()
 NA

Member [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_General](#) ()
 NA

Member [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_No_Detection](#) ()
 NA

Member [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_One_Detection](#) ()
 NA

Member [f360_statistical_tests_Check_Stationary_Hypothesis_handles_std_equal_to_zero](#) ()
 NA.

Member [f360_statistical_tests_Check_Stationary_Hypothesis_sigma_above_threshold_test_variable_below_threshold](#)
 ()
 NA.

Member [f360_statistical_tests_Check_Stationary_Hypothesis_sigma_and_test_variable_above_thresholds](#)
 ()
 NA.

Member [f360_statistical_tests_Check_Stationary_Hypothesis_sigma_below_threshold_test_variable_above_threshold](#)
 ()
 NA.

Member [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value_p_value_properly_calculated](#)
 ()
 NA.

Member [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value_p_value_properly_saturated](#)
 ()
 NA.

Member [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value_test_variable_greater_than_max_test_variab](#)
 ()
 NA.

Member [f360_statistical_tests_F360_Try_To_Reject_Test_properly_does_not_reject_hypothesis](#) ()
 NA.

Member [f360_statistical_tests_F360_Try_To_Reject_Test_properly_rejects_hypothesis](#) ()
 NA.

Member [f360_uncertainty_propagation_Test_Constant_Uncertainty_Propagation_2d](#) ()
 NA

Member [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_All](#) ()
NA

Member [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_All](#) ()
NA

Member [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#) ()
NA

Member [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#) ()
NA

Member [f360_update_sensor_first_detection_index_Update_Sensor_First_Detection_Index](#) ()
NA

Member [f360_update_sensor_first_detection_index_update_sensor_first_detection_index_too_many_detections](#) ()
NA

Member [f360_update_sensors_fov_update_sensors_fov](#) ()
NA

Member [f360_update_sensors_fov_update_sensors_fov_azimuth_beyond_fov](#) ()
NA

Member [f360_update_sensors_FunctionToTest_Descriptive_Tag](#) ()
Put in requirement tag if any otherwise set to NA.

Member [f360_update_sensors_valid_info](#) ()
: NA

Member [f360_update_sensors_valid_info_check_esr_valid](#) ()
: NA

Member [f360_update_sensors_valid_info_check_invalid_sensor](#) ()
: NA

Member [f360_update_sensors_valid_info_check_srr_esr_valid](#) ()
: NA

Member [f360_update_sensors_valid_info_check_srr_valid](#) ()
: NA

Member [IGNORE_TEST](#) (f360_ego_motion_compensate_dets__moving_forward_host_qualtest, Detection_Motion_Classification__det_rr_comp_calculation)
FTCP-8402

Member [IGNORE_TEST](#) (f360_ego_motion_compensate_dets__stopped_host_qualtest, Ego_Motion__Compensate_Dets__det_rr_comp)
FTCP-8402

Member [inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements](#) ()
FTCP-8256, FTCP-7799

Member [rspp_calibrations_FunctionToTest_Descriptive_Tag](#) ()
Put in requirement tag if any otherwise set to NA.

Member [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#) ()
FTCP-8841

Member [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch](#) ()
FTCP-8841

Member [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch \(\)](#)
FTCP-8841

Member [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true \(\)](#)
FTCP-8841

Member [rspp_detection_double_bounce_detector_Test_one_detection_per_sensor \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_inside_gate \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_1 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_2 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_3 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_1 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_2 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_3 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_4 \(\)](#)

Member [rspp_detection_double_bounce_detector_Test_with_2_detection_with_double_bounce \(\)](#)

Member [rspp_discard_detections_outside_sensor_fov_All_Detections_But_One_Outside_FoV \(\)](#)
NA

Member [rspp_discard_detections_outside_sensor_fov_All_Detections_Inside_FoV \(\)](#)
NA

Member [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_As_Per_Sensor_Lookid \(\)](#)
NA

Member [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_Based_On_Their_Sensor_Id \(\)](#)
NA

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Detection \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Sensor_Id \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_No_Detection \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Sensor_Valid \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_High_Interval_Width \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_Low_Interval_Width \(\)](#)
NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Two_Detections](#) ()

NA.

Member [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Valid_Detection](#) ()

NA.

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Base_Interval](#) ()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Both_In](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit](#)
()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Exactly_On_Limit](#) ()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_Base](#)

()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_First_Upper](#)

()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Second_Lower_Interval](#) ()

NA

Member [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Third_Upper_Interval](#) ()

NA

Member [system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario](#) ()

NA

Member [system_test_rspp_detection_double_bounce_detector_Double_Bounce_True](#) ()

NA

Member [system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True](#) ()

NA

Chapter 2

Module Documentation

2.1 Rspp_calibrations

Functions

- test group [rspp_calibrations](#) ()

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

- [rspp_calibrations_FunctionToTest_Descriptive_Tag](#) ()

2.1.1 Detailed Description

2.1.2 Function Documentation

2.1.2.1 rspp_calibrations()

```
test group rspp_calibrations ( )
```

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

Teardown:\n

Describe what is done in test teardown. Remove test teardown function and this tag if it is not used.

2.1.2.2 rspp_calibrations_FunctionToTest_Descriptive_Tag()

```
rspp_calibrations_FunctionToTest_Descriptive_Tag ( )
```

Purpose:\n

Describe purpose of test on a high level. The description should be unique and distinguishable from other tests' purposes. E.g. test that function Some_Function() works as intended given some specific preconditions.

Requirements Put in requirement tag if any otherwise set to NA.

Precondition:\n

Describe prepared input. E.g. describe relevant set up from test group and what variables are changed or added for this specific test.

Action:\n

Describe the action of the test. E.g. call Some_Function().

Result:\n

Describe expected output. E.g. check that the output match expected data.

2.2 F360_check_det_az_conf_and_elevation_qualtest

Functions

- test group [f360_check_det_az_conf_and_elevation_qualtest](#) ()
This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off](#) ()

2.2.1 Detailed Description

2.2.2 Function Documentation

2.2.2.1 f360_check_det_az_conf_and_elevation_qualtest()

```
test group f360_check_det_az_conf_and_elevation_qualtest ( )
```

This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.2.2.2 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation()

```
f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation ( )
```

Purpose:\nIf the confidence value is too low (3 being the lowest) or the

elevation angle is suspiciously high, do not allow detection to start a new track.

Requirements FTCP-8842

Step 1:

Generate Azimuth confidence level and elevation angle for MRR3 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

confidence value is too low or the elevation angle is high,new track is not started.

confidence value is lower boundary value and the elevation angle is low, new track is not started.

confidence value is lower boundary value or the elevation angle is too high,new track is not started.

Action:

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:

Check if the confidence level and elevation angle are in range and the results are as expected

2.2.2.3 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1 ()

Purpose: Generate Azimuth confidence level and elevation angle for SRR4 sensor detections (Need to modify the test case once there is requirement update)

Requirements FTCP-8842**Step 1:**

Generate Azimuth confidence level and elevation angle for SRR4 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated SRR4 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR4 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR4 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated SRR4 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated SRR4 sensor valid detections based on f_super_res and azimuth

Action:

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:

Check if the confidence level and elevation angle are in range and the results are as expected

2.2.2.4 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2 ()

Purpose: Generate Azimuth confidence level and elevation angle for SRR5 seonsor detections (Need to modify the test case once there is requireme

Requirements FTCP-8842

Step 1:

Generate Azimuth confidence level and elevation angle for SRR5 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated SRR5 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR5 sensor detections that are suspicious based on elevation angle

Check for unassociated SRR5 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated SRR5 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated SRR5 sensor valid detections based on f_super_res and azimuth

Action:

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:

Check if the confidence level and elevation angle are in range and the results are as expected

2.2.2.5 f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3()

f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3 ()

Purpose: Generate Azimuth confidence level and elevation angle for MRR360 seonsor detections (Need to modify the test case once there is requirement update)

Requirements FTCP-8842

Step 1:

Generate Azimuth confidence level and elevation angle for F360 seonsor detections (Need to modify the test case once there is requirement update)

Precondition:

Set up the detection list, sensor calibration, Host list and the expected detection properties.

Initialize local variables

Check for unassociated MRR360 sensor detections that are suspicious based on elevation angle

Check for unassociated MRR360 sensor detections that are suspicious based on elevation angle

Check for unassociated MRR360 sensor detections that are suspicious based on range, azimuth confidence

Check for unassociated MRR360 sensor detections that are suspicious based on range, azimuth, confidence value

Check for associated MRR360 sensor valid detections based on f_super_res and azimuth

Action:

Call Check_Detection_Azimuth_Confidence_and_Elevation() to check Azimuth confidence level and elevation angle.

Result:

Check if the confidence level and elevation angle are in range and the results are as expected

2.2.2.6 f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off()

f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off ()

Purpose:\n Test that the the elevation check is not performed if the f_mrr360_filter_away_big_elev_angle calib is false

Requirements FTCP-8842**Step 1:**\n

Initialize default parameters

Precondition:\n

Set host speed to less than min_host_speed so that the first check is not performed.

Create 5 detection with raw_detection_list->processed.f_ok_to_use = true

1. Sensor type = MRR3, $-6 < \text{elevation angle} < 6$ deg (0.104719755119660 rad)
2. Sensor type = MRR3, elevation angle > 6 deg (0.104719755119660 rad)
3. Sensor type = SRR5, elevation angle > 6 deg (0.104719755119660 rad)
4. Sensor type = MRR3, elevation angle < -6 deg (-0.104719755119660 rad)
5. Sensor type = SRR5, elevation angle < -6 deg (-0.104719755119660 rad)

Action:\n

Run Check_Detection_Azimuth_Confidence_and_Elevation

Result:\n

Check that all the detections are still ok to use (even the MRR360 with high elevation)

2.3 F360_check_det_az_conf_and_elevation

Functions

- test group `f360_check_det_az_conf_and_elevation` ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- `f360_check_det_az_conf_and_elevation_FunctionToTest_Descriptive_Tag` ()

2.3.1 Detailed Description

2.3.2 Function Documentation

2.3.2.1 `f360_check_det_az_conf_and_elevation()`

```
test group f360_check_det_az_conf_and_elevation ( )
```

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

Teardown:\n

Describe what is done in test teardown. Remove test teardown function and this tag if it is not used.

2.3.2.2 `f360_check_det_az_conf_and_elevation_FunctionToTest_Descriptive_Tag()`

```
f360_check_det_az_conf_and_elevation_FunctionToTest_Descriptive_Tag ( )
```

Purpose:\n

Describe purpose of test on a high level. The description should be unique and distinguishable from other tests' purposes. E.g. test that function `Some_Function()` works as intended given some specific preconditions.

Requirements Put in requirement tag if any otherwise set to NA.

Precondition:\n

Describe prepared input. E.g. describe relevant set up from test group and what variables are changed or added for this specific test.

Action:\n

Describe the action of the test. E.g. call `Some_Function()`.

Result:\n

Describe expected output. E.g. check that the output match expected data.

2.4 Rspp_compute_detections_properties

Functions

- test group [compute_detection_properties](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [compute_detection_properties_Compute_Detection_VCS_Coordinates](#) ()
- [compute_detection_properties_Test_Updated_detection_Time_with_data](#) ()
- [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_positive_polarity](#) ()
- [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_negative_polarity](#) ()

2.4.1 Detailed Description

2.4.2 Function Documentation

2.4.2.1 [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_negative_polarity](#)

```
compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_↵
Angle_negative_polarity ( )
```

Purpose:\n

Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle for positive polarity and verify they match the expected values

Requirements NA.

Precondition:\n

Set appropriate values needed as inputs to Compensate Detection VCS Azimuth And Elevation By The Misalignment_Angle

Action:\n

call [Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle](#)().

Result:\n

check that the output match expected data.

2.4.2.2 compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_positi

```
compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_↵  
Angle_positive_polarity ( )
```

Purpose:\n

Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle for positive polarity and verify they match the expected values

Requirements NA.

Precondition:\n

Set appropriate values needed as inputs to Compensate Detection VCS Azimuth And Elevation By The Misalignment_Angle

Action:\n

call Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle().

Result:\n

check that the output match expected data.

2.4.2.3 compute_detection_properties_Compute_Detection_VCS_Coordinates()

```
compute_detection_properties_Compute_Detection_VCS_Coordinates ( )
```

Purpose:\n

Compute_Detection_VCS_Coordinates along x y and z axes and verify they match the expected values

Requirements NA.

Precondition:\n

Set appropriate values needed as inputs to calculate detection vcs coordinates in x y and z axis

Action:\n

call Compute_Detection_VCS_Coordinates().

Result:\n

check that the output match expected data.

2.4.2.4 compute_detection_properties_Test_Updated_detection_Time_with_data()

```
compute_detection_properties_Test_Updated_detection_Time_with_data ( )
```

Purpose: Test Update_Detection_Property function and check on Time and global ID

Requirements NA

Precondition:

Test Update_Detection_Property with valid data

Action:

Call Update_Detection_Property

Result:

Check the expected value for cos_vcs_az, vcs_az, sin_vcs_az, range_rate_dealiased, rng_rate_interval_width, time_since_measurement, look_type and range_type

2.5 Rspg_detection_double_bounce_detector_qualtest

Functions

- test group [rspg_detection_double_bounce_detector_qualtest](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [rspg_detection_double_bounce_detector_qualtest_DoubleBounce_true](#) ()
- [rspg_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch](#) ()
- [rspg_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#) ()
- [rspg_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch](#) ()

2.5.1 Detailed Description

2.5.2 Function Documentation

2.5.2.1 [rspg_detection_double_bounce_detector_qualtest](#)()

```
test_group rspg_detection_double_bounce_detector_qualtest ( )
```

Basic test to verify that detection gets flagged as double bounce when expected and vice versa.

Setup:\n

Setup common data for all tests related to this test group

Teardown:\n

Nothing to teardown in this test group

2.5.2.2 [rspg_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#)()

```
rspg_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since azimuth condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.5.2.3 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since range condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.5.2.4 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch ( )
```

Purpose:\n Test that detection is NOT flagged as double bounce since range rate condition is not met

Requirements FTCP-8841

Result:\n

We expect no detection to be flagged as double bounce

2.5.2.5 rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true()

```
rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true ( )
```

Purpose:\n Test that detection is flagged as double bounce

Requirements FTCP-8841

Result:\n

We only expect the detection with higher range to be flagged as double bounce

2.6 System_test_rspp_detection_double_bounce_detector

Functions

- test group `system_test_rspp_detection_double_bounce_detector` ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- `system_test_rspp_detection_double_bounce_detector_Double_Bounce_True` ()
- `system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True` ()
- `system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario` ()

2.6.1 Detailed Description

2.6.2 Function Documentation

2.6.2.1 system_test_rspp_detection_double_bounce_detector()

```
test_group system_test_rspp_detection_double_bounce_detector ( )
```

Basic test to verify that detection gets flagged as double bounce when expected and vice versa.

Setup:\n

Setup common data for all tests related to this test group

Setup 4 detections in a double bounce pattern. Modify these values and maybe add more detections to define test cases. The following detections are setup:

det[x]	.range	.range_rate	.azimuth	.sensor_id
det[0]	10.0	2.0	0.1	1
det[1]	20.0	4.0	0.11	1
det[2]	30.0	6.0	0.11	1
det[3]	40.0	8.0	0.11	1

Teardown:\n

Nothing to teardown in this test group

2.6.2.2 system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario()

```
system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario ( )
```


Purpose:\n Test Arbitrary Scenario 1

Requirements NA

Step 1:\n

Testing arbitrary scenario

Precondition:\n

Creating new set of 7 detections:

2.6.2.3 -----X6-----

2.6.2.4 -----O3---X4-----x5-----

2.6.2.5 -----O0-O1-O2-----

-----S0-----

Character	Type
X	Multibounce detection
O	Real detection
S	Sensor

Action:\n

Call the function

Result:\n

We only expect the detection with higher range to be flagged as double bounce

2.6.2.6 system_test_rspp_detection_double_bounce_detector_Double_Bounce_True()

system_test_rspp_detection_double_bounce_detector_Double_Bounce_True ()

Purpose:\n Simple test using two detections only, one is double bounce of the other one

Requirements NA

Step 1:

Testing 2 detections

Precondition:

Testing only 2 detections of the set

Action:

Call the function

Result:

We only expect the detection with higher range to be flagged as double bounce

2.6.2.7 system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True()

```
system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True ( )
```

Purpose: Checking multiple bounces, one reference detection have yielded 3 double bounces

Requirements NA

Step 1:

Testing multiple bounces

Precondition:

Testing 4 detections of the set

Action:

Call the function

Result:

We only expect the detection with higher range to be flagged as double bounce

2.7 Rspp_detection_double_bounce_detector

Functions

- test group `rspp_detection_double_bounce_detector` ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- `rspp_detection_double_bounce_detector_Test_with_2_detection_with_double_bounce` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_4` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_3` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_2` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_1` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_inside_gate` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_3` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_2` ()
- `rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_1` ()
- `rspp_detection_double_bounce_detector_Test_one_detection_per_sensor` ()

2.7.1 Detailed Description

2.7.2 Function Documentation

2.7.2.1 `rspp_detection_double_bounce_detector_Test_one_detection_per_sensor()`

`rspp_detection_double_bounce_detector_Test_one_detection_per_sensor` ()

Purpose:\n Test Double_Bounce_Detection_Countermeasure with only one detection per sensor

Requirements

Precondition:\n

Set up 4 detections with different sensor IDs.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.2 rspp_detection_double_bounce_detector_Test_with_2_detection_inside_gate()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_inside_gate ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection inside Gate

Requirements

Precondition:\n

Set up 2 detections inside the gate by changing the calibration values.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that the secondary detection is flagged as double bounce and other detections are not flagged as double bounce.

2.7.2.3 rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_1()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_1 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection not in limit

Requirements

Precondition:\n

Set up 2 detections not in the limit: range is 0.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.4 rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_2()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_2 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection not in limit

Requirements

Precondition:\n

Set up 2 detections not in the limit: number of multi-bounce is less than 1 (The object range is higher than double the limit).

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.5 rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_3()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_3 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection not in limit

Requirements

Precondition:\n

Set up 2 detections not in the limit: max number of multi-bounce is less than the number of multi-bounce.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.6 rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_1()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_1 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection outside gate for being in cone of silence

Requirements

Precondition:\n

Set up 2 detections outside the gate: vcs azimuth for the primary detection is out of border.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.7 rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_2()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_2 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection outside gate for being in cone of silence

Requirements

Precondition:\n

Set up 2 detections outside the gate: vcs azimuth for the primary detection is out of border.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.8 rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_3()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_3 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection outside gate for being in cone of silence

Requirements

Precondition:\n

Set up 2 detections outside the gate: The primary detection is less than range_rate_threshold.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.9 rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_4()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_4 ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection outside gate for being in cone of silence

Requirements

Precondition:\n

Set up 2 detections outside the gate: vcs azimuth for the secondary detection is out of border.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that all detections are not flagged as double bounce.

2.7.2.10 rspp_detection_double_bounce_detector_Test_with_2_detection_with_double_bounce()

```
rspp_detection_double_bounce_detector_Test_with_2_detection_with_double_bounce ( )
```

Purpose:\n Test Double_Bounce_Detection_Countermeasure with 2 detection with double bounce in the cone of silence

Requirements

Precondition:\n

Set up the 2 detection to be bounce in the cone of silence.

Action:\n

Call Double_Bounce_Detection_Countermeasure.

Result:\n

Check that secondary detection is double bounce and other detections is not double bounce

2.8 F360_detection_multipath_filter

Functions

- test group [f360_detection_multipath_filter](#) ()
Test that the functionality in Filter_Out_Multipath_Detections() works as intended.
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_First_Det_Idx_Invalid](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Above_Thresh](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Below_Thresh](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Num_Valid_Dets_Below_Threshold](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Host_Slow](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter__Not_SRR5](#) ()

2.8.1 Detailed Description

2.8.2 Function Documentation

2.8.2.1 f360_detection_multipath_filter()

```
test group f360_detection_multipath_filter ( )
```

Test that the functionality in Filter_Out_Multipath_Detections() works as intended.

Setup:\n

Set up a correct calibration structure and specify fields so that the filter is activated and modifies the detection property f_ok_to_use.

2.8.2.2 f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_First_Det_Idx_Invalid()

```
f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_First_Det_Idx_Invalid ( )
```

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended and doesn't mark the detections as not ok to use when the first_detection_list_idx in sensor props is invalid for the given sensor.

Requirements NA

Precondition:\n

Set first detection index in sensor props to RSPP_INVALID_ID

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detections are not marked as not ok to use.

2.8.2.3 f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Above_Thresh()

```
f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Above_Thresh (
)
```

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended and marks the detections as not ok to use when the fraction is higher than the threshold for an SRR5 sensor.

Requirements NA

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detections with confid_az == RSPP_CONF_AZIMUTH_LOW are marked as not ok to use and that those with higher azimuth confidence are still ok to use.

2.8.2.4 f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Below_Thresh()

```
f360_detection_multipath_filter_Detection_Multipath_Filter__Filter_Num_Bad_Dets_Below_Thresh (
)
```

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended and doesn't mark the detections as not ok to use when the fraction is below than the threshold for an SRR5 sensor.

Requirements NA

Precondition:\n

Change azimuth confidence of one detection from RSPP_CONF_AZIMUTH_LOW to RSPP_CONF_AZIMUTH_HIGH such that the fraction of dets with bad azimuth is slightly below threshold.

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detections are not marked as not ok to use.

2.8.2.5 f360_detection_multipath_filter_Detection_Multipath_Filter__Host_Slow()

f360_detection_multipath_filter_Detection_Multipath_Filter__Host_Slow ()

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended when host is driving too slow to activate the filter.

Requirements NA

Precondition:\n

Set host speed to something below calibration value 1.0 m/s.

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detection is still marked as ok to use.

2.8.2.6 f360_detection_multipath_filter_Detection_Multipath_Filter__Not_SRR5()

f360_detection_multipath_filter_Detection_Multipath_Filter__Not_SRR5 ()

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended and marks the detection as ok to use when the sensor is of a different type than SRR5.

Requirements NA

Precondition:\n

Define array of possible sensor types (where SRR5 is excluded).

Precondition:\n

Set sensor type.

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detection is still marked as ok to use.

2.8.2.7 f360_detection_multipath_filter_Detection_Multipath_Filter__Num_Valid_Dets_Below_Threshold()

```
f360_detection_multipath_filter_Detection_Multipath_Filter__Num_Valid_Dets_Below_Threshold ( )
```

Purpose:\n

Test that function Filter_Out_Multipath_Detections() works as intended when there are not enough valid detections present for the sensor.

Requirements NA

Precondition:\n

Set number of valid detections for the sensor to 50.

Action:\n

Call Filter_Out_Multipath_Detections()

Result:\n

Check that the detection is still marked as ok to use.

2.9 Rspp_discard_detections_outside_sensor_fov

Functions

- test group [rspp_discard_detections_outside_sensor_fov](#) ()
This test group verifies the functionality of Discard_Detections_Outside_Sensor_Field_Of_View.
- [rspp_discard_detections_outside_sensor_fov_All_Detections_Inside_FoV](#) ()
- [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_Based_On_Their_Sensor_Id](#) ()
- [rspp_discard_detections_outside_sensor_fov_All_Detections_But_One_Outside_FoV](#) ()
- [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_As_Per_Sensor_Lookid](#) ()

2.9.1 Detailed Description

2.9.2 Function Documentation

2.9.2.1 rspp_discard_detections_outside_sensor_fov()

```
test_group rspp_discard_detections_outside_sensor_fov ( )
```

This test group verifies the functionality of Discard_Detections_Outside_Sensor_Field_Of_View.

Setup:\n

Set up 2 sensors with 2, different, field of view Set up 5 detections to be ok to use, originate from the sensor 1, and have azimuth and elevation that is inside sensor id 1 FoV

2.9.2.2 rspp_discard_detections_outside_sensor_fov_All_Detections_But_One_Outside_FoV()

```
rspp_discard_detections_outside_sensor_fov_All_Detections_But_One_Outside_FoV ( )
```

Purpose:\n

Verify detections outside the sensor field of view are flagged as not ok to use

Requirements NA

Precondition:\n

Set 2 detections to be outside the sensor FoV, let 1 be inside

Action:\n

Call Discard_Detections_Outside_Sensor_Field_Of_View()

Result:\n

Compare against expected data

2.9.2.3 rspp_discard_detections_outside_sensor_fov_All_Detections_Inside_FoV()

```
rspp_discard_detections_outside_sensor_fov_All_Detections_Inside_FoV ( )
```

Purpose:\n

Verify detections inside the sensor field of view are not flagged

Requirements NA

Precondition:\n

Set all detections to be inside sensorid 1 lookid0 FoV

Action:\n

Call Discard_Detections_Outside_Sensor_Field_Of_View()

Result:\n

Compare against expected data

2.9.2.4 rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_As_Per_Sensor_Lookid()

```
rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_As_Per_Sensor_Lookid ( )
```

Purpose:\n

Verify detections outside the sensor field of view are flagged based on the sensors' look id

Requirements NA

Precondition:\n

Set number of valid detection to 2 Set detection 0 to be inside sensor FoV for lookid 0 Set detection 1 to be inside sensor FoV for lookid 3 Set sensor lookid to 3

Action:\n

Call Discard_Detections_Outside_Sensor_Field_Of_View()

Result:\n

Compare against expected data

2.9.2.5 rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_Based_On_Their_Sensor_Id()

```
rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_Based_On_Their_↵  
Sensor_Id ( )
```

Purpose:

Verify detections are flagged based on their sensor id

Requirements NA**Precondition:**

Set number of valid detection to 3 Set detection 0 to be inside the sensorid 1 lookid 1 FoV Set detection 1 to be inside the sensorid 1 lookid 3 FoV (i.e. outside sensorid 1 current lookid) Set detection 2 to be inside the sensorid 2 lookid 0 FoV

Action:

Call Discard_Detections_Outside_Sensor_Field_Of_View()

Result:

Compare against expected data

2.10 F360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host](#) ()

Check correctness of detection motion status determination when host does not move.

- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_low](#)
()

2.10.1 Detailed Description

2.10.2 Function Documentation

2.10.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host()

```
test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host ( )
```

Check correctness of detection motion status determination when host does not move.

Setup:\n

- Init calibs.
- Setup one detection.

2.10.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_amb_due_to_being_az_error_stat ( )
```

Purpose:\n

Check correctness of detection motion determination when detections is marked as f_azimuth_error_stat_mov

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.10.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_amb_due_to_low_conf_az ( )
```

Purpose:\n

Check correctness of detection motion determination when det's conf az is low

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.10.2.4 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_1 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to 0.0F

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.10.2.5 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_2 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to 0.0F

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.10.2.6 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_ambiguous_dealiased_case_3 ( )
```

Purpose:

Check correctness of detection motion determination when det's range rate is high

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as ambiguous

2.10.2.7 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_det_as_moving_dealiased_case_1 ( )
```

Purpose:

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as moving

2.10.2.8 f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification_

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification↔  
__mark_low_range_rate_moving ( )
```

Purpose:

Check correctness of detection motion determination when detections is marked as f_azimuth_error_stat_mov

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as ambiguous

2.11 F360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host

Functions

- test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host` ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_1` ()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_2` ()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_3` ()

2.11.1 Detailed Description

2.11.2 Function Documentation

2.11.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host()

test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host` ()

Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)

Setup:\n

- Init calibs.
- Setup one detection.

2.11.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_1()

`f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__mark_det_as_ambiguous_dealiased_case_1` ()

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to -10.0F and relative high motion threshold

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call `Compute_Detections_Properties_Based_On_Ego_Motion()`.

Result:\n

Detection marked as ambiguous

2.11.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classi

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion↔  
_Classification__mark_det_as_ambiguous_dealiased_case_2 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is close to -10.0F and relative low motion threshold

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.11.2.4 f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classi

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion↔  
_Classification__mark_det_as_moving_dealiased_case_1 ( )
```

Purpose:\n

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190 NA.

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.12 F360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host ↩

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host](#) ()
Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#) ()

2.12.1 Detailed Description

2.12.2 Function Documentation

2.12.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host()

```
test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host (
)
```

Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.

Setup:\n

- Init calibs.
- Setup one detection.

2.12.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_C

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_↩
Motion_Classification__aliased_amb_det ( )
```

Purpose:\n

Check correctness of detection motion determiantion when det's range rate is too high

Requirements FTCP-8190

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection marked as ambiguous

2.12.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_C

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_↔  
Motion_Classification__aliased_moving_det ( )
```

Purpose:

Check correctness of detection motion determination when det's range rate is high enough

Requirements FTCP-8190

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection marked as moving

2.13 F360_Compute_Detections_Properties_Based_On_Ego_Motion__f_esr_mr_in_lr_fov ↩

Functions

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__f_esr_mr_in_lr_fov](#) ()
Check if f_esr_mr_in_lr_fov is calculated correctly.

2.13.1 Detailed Description

2.13.2 Function Documentation

2.13.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__f_esr_mr_in_lr_fov()

```
test group f360_Compute_Detections_Properties_Based_On_Ego_Motion__f_esr_mr_in_lr_fov ( )
```

Check if f_esr_mr_in_lr_fov is calculated correctly.

Setup:\n

- Init calibs.
- Setup one detection.

2.14 F360_Compute_Detections_Properties_Based_On_Ego_Motion__↔ threshold_limitation

Functions

- test group `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation` ()
Check if the threshold limitation introduced to handle low speed crossing vehicles is working as intended. Detections need to be able to get classified as moving even with low compensated range-rate in order to have acceptable tracking. However, when the host is moving a high speed or during high curvature the threshold should remain high.
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_within_inte`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_below_inte`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_above_inte`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_if_infinite`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_at_upper`
()
- `f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_at_lower`
()

2.14.1 Detailed Description

2.14.2 Function Documentation

2.14.2.1 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation()

```
test_group f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation ( )
```

Check if the threshold limitation introduced to handle low speed crossing vehicles is working as intended. Detections need to be able to get classified as moving even with low compensated range-rate in order to have acceptable tracking. However, when the host is moving a high speed or during high curvature the threshold should remain high.

Setup:\n

- Init calibs.

2.14.2.2 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←
_rate_comp_above_interval_width ( )
```

Purpose:

Test to check that the range_rate_compensated is adjusted accordingly when it lies above the stationary hypothesis bounds $[-0.5*v_wrapping, 0.5*v_wrapping]$

Requirements NA.

Precondition:

Set appropriate data such that calculated range rate compensated should be 50 and the v_wrapping are $[-16, 16]$

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

For interval widths between $[-16, 16]$ expected_range_rate_compensated value should be -14, for range_rate←_compensated = 50

2.14.2.3 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←
_rate_comp_at_lower_bound ( )
```

Purpose:

Test to check that the range_rate_compensated is returned unmodified when it equals the lower stationary hypothesis bounds $[-0.5*v_wrapping, 0.5*v_wrapping]$

Requirements NA.

Precondition:

Set appropriate data such that calculated range rate compensated should be -16 and the v_wrapping are $[-16, 16]$

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

For interval widths between $[-16, 16]$ expected_range_rate_compensated value should be -16, for range_rate←_compensated= -16

2.14.2.4 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←
_rate_comp_at_upper_bound ( )
```

Purpose:\n

Test to check that the range_rate_compensated is returned unmodified when it equals the upper stationary hypothesis bounds $[-0.5*v_wrapping, 0.5*v_wrapping]$

Requirements NA.

Precondition:\n

Set appropriate data such that calculated range rate compensated should be 16 and the v_wrapping are [-16,16]

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

For interval widths between [-16 16]expected_range_rate_compensated value should be 16, for range_rate←_compensated= 16

2.14.2.5 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←
_rate_comp_below_interval_width ( )
```

Purpose:\n

Test to check that the range_rate_compensated is adjusted accordingly when it lies below the stationary hypothesis bounds $[-0.5*v_wrapping, 0.5*v_wrapping]$

Requirements NA.

Precondition:\n

Set appropriate data such that calculated range rate compensated should be -40 and the v_wrapping are [-16,16]

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

For interval widths between [-16 16]expected_range_rate_compensated value should be -8, for range_rate←_compensated= -40

2.14.2.6 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←  
_rate_comp_if_infinite ( )
```

Purpose:

Test to check that the range_rate_compensated is reported even for extremely large values without being stuck

Requirements NA.

Precondition:

Set appropriate data such that calculated range rate compensated is very large

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Expect a large compensated value without being stuck

2.14.2.7 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range←  
_rate_comp_within_interval_width ( )
```

Purpose:

Test to check that the range_rate_compensated is returned unmodified when it lies between the stationary hypothesis bounds $[-0.5*v_wrapping, 0.5*v_wrapping]$

Requirements NA.

Precondition:

Set appropriate data such that calculated range_rate_compensated should be 10 and the v_wrapping are [-16,16]

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

expected_range_rate_compensated value should be 10, since the value lies in the interval widths [-16 16]

2.14.2.8 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Prop

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections↔  
_Properties_Based_On_Ego_Motion__high_speed_low_curvature ( )
```

Purpose:\n

Check that the threshold is high when host speed is high and curvature is low

Requirements NA.

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection is classified ambiguous

2.14.2.9 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Prop

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections↔  
_Properties_Based_On_Ego_Motion__low_azimuth_conf ( )
```

Purpose:\n

Check that the threshold is high / bypassed when azimuth confidence is low for a detection range > 10m

Requirements NA.

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection is classified ambiguous

2.14.2.10 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Pro

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections↔  
_Properties_Based_On_Ego_Motion__low_speed_high_curvature ( )
```

Purpose:

Check that the threshold is high when host speed is low and curvature is high

Requirements NA.

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection is classified ambiguous

2.14.2.11 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Pro

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections↔  
_Properties_Based_On_Ego_Motion__low_threshold ( )
```

Purpose:

Check that the threshold is low when host speed and curvature is low

Requirements NA.

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection is classified moving

2.14.2.12 f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Pro

```
f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections↵  
_Properties_Based_On_Ego_Motion__low_threshold_reversing_host ( )
```

Purpose:

Check that the threshold is low when host speed and curvature is low but the host is reversing

Requirements NA.

Precondition:

NA

Action:

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:

Detection is classified moving

2.15 Rspp_ego_motion_compensate_dets_stopped_host_qualtest

Collaboration diagram for Rspp_ego_motion_compensate_dets_stopped_host_qualtest:

2.16 F360_ego_motion_compensate_dets__moving_forward_host_↔ qualtest

Collaboration diagram for F360_ego_motion_compensate_dets__moving_forward_host_qualtest:

Functions

- test group [f360_ego_motion_compensate_dets__moving_forward_host_qualtest](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [IGNORE_TEST](#) ([f360_ego_motion_compensate_dets__moving_forward_host_qualtest](#), [Detection_Motion↔_Classification__det_rr_comp_calculation](#))
- [f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_comp_calculation](#) ()

2.16.1 Detailed Description

2.16.2 Function Documentation

2.16.2.1 f360_ego_motion_compensate_dets__moving_forward_host_qualtest()

```
test_group f360_ego_motion_compensate_dets__moving_forward_host_qualtest ( )
```

Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)

Setup:\n

- Init calibs.
- Setup one detection.

Setup:\n

- Init calibs.
- Setup one detection.

2.16.2.2 f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_c

```
f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification↔
__det_rr_comp_calculation ( )
```

Purpose:\n

Check correctness of detection range rate compensatoad calculation

Requirements FTCP-8402

Precondition:\n

NA

Action:\n

Call Internal_Preprocessing().

Result:\n

Detection range_rate_compensated is equal to 20.0F

2.16.2.3 IGNORE_TEST()

```
IGNORE_TEST (
    f360_ego_motion_compensate_dets__moving_forward_host_qualtest ,
    Detection_Motion_Classification__det_rr_comp_calculation )
```

Purpose:\n

Check correctness of detection range rate compensatoad calculation

Requirements FTCP-8402

Precondition:\n

NA

Action:\n

Call Compute_Detections_Properties_Based_On_Ego_Motion().

Result:\n

Detection range_rate_compensated is equal to 20.0F

2.17 F360_filter_detections_with_bad_azimuth_confidence

Functions

- test group `f360_filter_detections_with_bad_azimuth_confidence` ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- `f360_filter_detections_with_bad_azimuth_confidence_FunctionToTest_Descriptive_Tag` ()

2.17.1 Detailed Description

2.17.2 Function Documentation

2.17.2.1 f360_filter_detections_with_bad_azimuth_confidence()

```
test group f360_filter_detections_with_bad_azimuth_confidence ( )
```

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

Teardown:\n

Describe what is done in test teardown. Remove test teardown function and this tag if it is not used.

2.17.2.2 f360_filter_detections_with_bad_azimuth_confidence_FunctionToTest_Descriptive_Tag()

```
f360_filter_detections_with_bad_azimuth_confidence_FunctionToTest_Descriptive_Tag ( )
```

Purpose:\n

Describe purpose of test on a high level. The description should be unique and distinguishable from other tests' purposes. E.g. test that function `Some_Function()` works as intended given some specific preconditions.

Requirements Put in requirement tag if any otherwise set to NA.

Precondition:\n

Describe prepared input. E.g. describe relevant set up from test group and what variables are changed or added for this specific test.

Action:\n

Describe the action of the test. E.g. call `Some_Function()`.

Result:\n

Describe expected output. E.g. check that the output match expected data.

2.18 Rspp_get_version

Functions

- test group [rspp_get_version](#) ()
Check that RSPP_Get_Version returns the expected values.
- [rspp_get_version_RSPP_Get_Version_Output_Test](#) ()

2.18.1 Detailed Description

2.18.2 Function Documentation

2.18.2.1 rspp_get_version_RSPP_Get_Version_Output_Test()

```
rspp_get_version_RSPP_Get_Version_Output_Test ( )
```

Purpose:\n

Check that RSPP_Get_Version returns the expected values.

Precondition:\n

Set up variables to populate with version info

Action:\n

call RSPP_Get_Version()

Result:\n

Check that the output match expected data.

2.19 F360_host_props_update

Functions

- test group [f360_host_props_update \(\)](#)
Test group for Host_Props_Update. Tests in this test group verify that the host properties are time updated correctly.
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Heading_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Covariance_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Position_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Heading_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Covariance_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Position_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Covariance_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Position_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Yawing_Heading_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Yawing_Covariance_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Yawing_Position_Update \(\)](#)
- [f360_host_props_update_Host_Props_Update_Yawing_Heading_Normalized \(\)](#)

2.19.1 Detailed Description

2.19.2 Function Documentation

2.19.2.1 f360_host_props_update()

test_group f360_host_props_update ()

Test group for Host_Props_Update. Tests in this test group verify that the host properties are time updated correctly.

Setup:\n

Set up a default scenario where host is driving straight along wcs coordinate system x axis.

2.19.2.2 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Covariance_Update()

f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Covariance_Update ()

Purpose:\n

Test that host velocity and position increment covariance matrices are updated correctly when host is driving straight along wcs x axis.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis with a speed of 2 m/s.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host velocity and position increment covariance matrices have been updated as expected.

2.19.2.3 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Heading_Update()

f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Heading_Update ()

Purpose:\n

Test that host heading and delta pointing is updated correctly when host is driving straight along wcs x axis.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host heading did not change and that host delta pointing is zero.

2.19.2.4 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Position_Update()

f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Position_Update ()

Purpose:\n

Test that host wcs position and delta wcs position are updated correctly when host is driving straight along wcs x axis.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis with a speed of 2 m/s.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host wcs position as well as delta wcs position have been updated as expected.

2.19.2.5 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Covariance_Update()

f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Covariance_Update ()

Purpose:\n

Test that host velocity and position increment covariance matrices are updated correctly when host is driving straight along wcs x axis.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight in test group. Set host heading to $\pi / 2$.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host velocity and position increment covariance matrices have been updated as expected.

2.19.2.6 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Heading_Update()

f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Heading_Update ()

Purpose:\n

Test that host heading and delta pointing is updated correctly when host is driving straight along wcs y axis.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight in test group. Set host heading to $\pi / 2$.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host heading did not change and that host delta pointing is zero.

2.19.2.7 f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Position_Update()

```
f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Position_Update ( )
```

Purpose:

Test that host wcs position and delta vcs position are updated correctly when host is driving straight along wcs y axis.

Requirements NA.

Precondition:

Host properties have been set up such that host is driving straight in test group. Set host heading to $\pi / 2$.

Action:

Call Host_Props_Update

Result:

Verify that host wcs position and delta vcs position have been updated as expected.

2.19.2.8 f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Covariance_Update()

```
f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Covariance_Update ( )
```

Purpose:

Test that host velocity and position increment covariance matrices are updated correctly when host has a slight sideslip.

Requirements NA.

Precondition:

Host properties have been set up such that host is driving straight along wcs x axis in test group. Set host vcs sideslip to 5 degrees

Action:

Call Host_Props_Update

Result:

Verify that host velocity and position increment covariance matrices have been updated as expected.

2.19.2.9 f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Position_Update()

f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Position_Update ()

Purpose:\n

Test that host wcs position and delta vcs position are updated correctly when host has a slight sideslip.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis in test group. Set host vcs sideslip to 5 degrees

Action:\n

Call Host_Props_Update

Result:\n

Verify that host wcs position as well as delta vcs position have been updated as expected.

2.19.2.10 f360_host_props_update_Host_Props_Update_Yawing_Covariance_Update()

f360_host_props_update_Host_Props_Update_Yawing_Covariance_Update ()

Purpose:\n

Test that host velocity and position increment covariance matrices are updated correctly when host is yawing.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis with a speed of 2 m/s. Set host yaw rate to 0.1.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host velocity and position increment covariance matrices have been updated as expected.

2.19.2.11 f360_host_props_update_Host_Props_Update_Yawing_Heading_Normalized()

f360_host_props_update_Host_Props_Update_Yawing_Heading_Normalized ()

Purpose:\n

Test that host heading is normalized correctly when heading exceeds 180 degrees.

Requirements NA.

Precondition:\n

Set host heading angle to 180 degrees. Set host yaw rate such that host delta pointing is 5 degrees (yaw rate * elapsed time = delta pointing).

Action:\n

Call Host_Props_Update

Result:\n

Verify that host heading and delta pointing have been updated as expected.

2.19.2.12 f360_host_props_update_Host_Props_Update_Yawing_Heading_Update()

f360_host_props_update_Host_Props_Update_Yawing_Heading_Update ()

Purpose:\n

Test that host heading and delta pointing is updated correctly when host is yawing.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis with a speed of 2 m/s. Set host yaw rate to 0.1.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host heading and delta pointing have been updated as expected.

2.19.2.13 f360_host_props_update_Host_Props_Update_Yawing_Position_Update()

f360_host_props_update_Host_Props_Update_Yawing_Position_Update ()

Purpose:\n

Test that host wcs position and delta vcs position are updated correctly when host is yawing.

Requirements NA.

Precondition:\n

Host properties have been set up such that host is driving straight along wcs x axis with a speed of 2 m/s. Set host yaw rate to 0.1.

Action:\n

Call Host_Props_Update

Result:\n

Verify that host wcs position as well as delta vcs position have been updated as expected.

2.20 Rspp_input_sanity_check

Functions

- test group `rspp_input_sanity_check ()`
Testing behaviour of the RSPP Input_Sanity_Check() function.
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Valid_Detection ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_No_Detection ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Two_Detections ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Detection ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Sensor_Id ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_Low_Interval_Width ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_High_Interval_Width ()`
- `rspp_input_sanity_check_Input_Sanity_Check_Test_With_Sensor_Valid ()`

2.20.1 Detailed Description

2.20.2 Function Documentation

2.20.2.1 rspp_input_sanity_check()

```
test group rspp_input_sanity_check ( )
```

Testing behaviour of the RSPP Input_Sanity_Check() function.

Setup:\n

Setup common valid inputs.

2.20.2.2 rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Detection()

```
rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Detection ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects faulty detection data and invalidate all detections when found. If input data is correct it shall remain untouched.

Requirements NA.

Precondition:\n

Setup invalid detection input data.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset when faulty detection and that sensors range rate interval also is intact.

2.20.2.3 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Invalid_Sensor_Id()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Invalid_Sensor_Id ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects faulty detection data and invalidate all detections when found. Range rate interval input data shall remain untouched.

Requirements NA.

Precondition:\n

Setup valid input data.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset when faulty detection and that sensors range rate interval also is intact.

2.20.2.4 rspp_input_sanity_check_Input_Sanity_Check__Test_With_No_Detection()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_No_Detection ( )
```

Purpose:\n

Test that function Input_Sanit_Check keeps input data untouched when no detections.

Requirements NA.

Precondition:\n

Setup input data with zero number of detections set.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset and range rate intervals are untouched.

2.20.2.5 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Sensor_Valid()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Sensor_Valid ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects faulty detection data and invalidate all detections when found. If input data is correct it shall remain untouched.

Requirements NA.

Precondition:\n

Setup valid input data but variable.is_valid = false.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset when faulty detection and that sensors range rate interval also is intact.

2.20.2.6 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Too_High_Interval_Width()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Too_High_Interval_Width ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects too high range rate interval widts value. If detected then default range rate interval width shall be set and number of detections set to zero.

Requirements NA.

Precondition:\n

Setup invalid input data.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have been reset and that sensors range rate interval also is set to default value.

2.20.2.7 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Too_Low_Interval_Width()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Too_Low_Interval_Width ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects too low range rate interval widths value. If detected then default range rate interval width shall be set and number of detections set to zero.

Requirements NA.

Precondition:\n

Setup invalid input data.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have been reset and that sensors range rate interval also is set to default value.

2.20.2.8 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Two_Detections()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Two_Detections ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects faulty detection data and invalidate all detections when found. If input data is correct it shall remain untouched.

Requirements NA.

Precondition:\n

Setup input data for 2 detections. First is invalid second is valid

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset when faulty detection and that sensors range rate interval also is intact.

2.20.2.9 rspp_input_sanity_check_Input_Sanity_Check__Test_With_Valid_Detection()

```
rspp_input_sanity_check_Input_Sanity_Check__Test_With_Valid_Detection ( )
```

Purpose:\n

Test that function Input_Sanit_Check detects faulty detection data and invalidate all detections when found. If input data is correct it shall remain untouched.

Requirements NA.

Precondition:\n

Setup valid input data done by TEST_SETUP.

Action:\n

Call Input_Sanity_Check function

Result:\n

Verify that number of detection have NOT been reset when faulty detection and that sensors range rate interval also is intact.

2.21 F360_ego_motion_compensate_dets__stopped_host_qualtest

Collaboration diagram for F360_ego_motion_compensate_dets__stopped_host_qualtest:

Modules

- [F360_ego_motion_compensate_dets__moving_forward_host_qualtest](#)

Functions

- test group [f360_ego_motion_compensate_dets__stopped_host_qualtest](#) ()
Check correctness of detection motion status determination when host does not move.
- [f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp](#) ()

2.21.1 Detailed Description

2.21.2 Function Documentation

2.21.2.1 f360_ego_motion_compensate_dets__stopped_host_qualtest()

```
test_group f360_ego_motion_compensate_dets__stopped_host_qualtest ( )
```

Check correctness of detection motion status determination when host does not move.

Setup:\n

- Init calibs.
- Setup one detection.

2.21.2.2 f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp()

```
f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp  
( )
```

Purpose:\n

Check if detection's range_rate_dealiased parameter is correctly filled

Requirements FTCP-8402

Precondition:\n

Setup detection raw values

Action:\n

Call Internal_Preprocessing().

Result:\n

Correct value of range_rate_compensated

2.22 F360_inputs_preprocessing

Functions

- test group [f360_inputs_preprocessing](#) ()
- [f360_inputs_preprocessing_Sensor_Capability_Host__SCM_Yaw_Rate_Std](#) ()

2.22.1 Detailed Description

2.22.2 Function Documentation

2.22.2.1 f360_inputs_preprocessing()

```
test_group f360_inputs_preprocessing ( )
```

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

2.22.2.2 f360_inputs_preprocessing_Sensor_Capability_Host__SCM_Yaw_Rate_Std()

```
f360_inputs_preprocessing_Sensor_Capability_Host__SCM_Yaw_Rate_Std ( )
```

Purpose:\n

Test that standard deviation for host yaw rate from SCM is updated correctly.

Requirements NA.

Precondition:\n

Default values with elapsed time equal to 0.05F

Action:\n

Call `Sensor_Capability_Host`

Result:\n

Verify that standard deviation for host yaw rate has been updated as expected.

2.23 F360_math_func

Functions

- test group [f360_math_func \(\)](#)
 - Tests for f360_math_func.*
 - [f360_math_func_test_f360_get_hypotenuse_negative_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_positive_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_zero_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_negative_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_positive_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_zero_input \(\)](#)
 - [f360_math_func_F360_Matmul_MxN_NxP \(\)](#)
 - [f360_math_func_F360_Matmul_MxN_NxP_Transpose \(\)](#)
 - [f360_math_func_F360_Saturate \(\)](#)
 - [f360_math_func_F360_Sort_fail \(\)](#)
 - [f360_math_func_F360_Sort_success \(\)](#)
 - [f360_math_func_F360_Sort_asc \(\)](#)
 - [f360_math_func_F360_Sort_sorted_values_not_modified \(\)](#)
 - [f360_math_func_F360_Sort_partial \(\)](#)
 - [f360_math_func_F360_Sort_no_sort \(\)](#)
 - [f360_math_func_F360_Sort_stability \(\)](#)
 - [f360_math_func_F360_Sort_dsc \(\)](#)
 - [f360_math_func_F360_Linear_Equation \(\)](#)
 - [f360_math_func_F360_Linear_Equation_small_den \(\)](#)
 - [f360_math_func_Test_F360_2d_Matrix_Determinant \(\)](#)
 - [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints \(\)](#)
 - [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints \(\)](#)
 - [f360_math_func_Test_F360_Linear_Equation_With_Saturation \(\)](#)
 - [f360_math_func_F360_Linear_Equation_With_Saturation_small_den \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_First_Order \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas \(\)](#)
 - [f360_math_func_Test_F360_Safe_Logf \(\)](#)
 - [f360_math_func_Test_F360_Power_Scale_Prob \(\)](#)
 - [f360_math_func_Test_F360_Accumulate_Probabilities \(\)](#)
 - [f360_math_func_Test_F360_matmul_2x2_2x1 \(\)](#)
 - [f360_math_func_Test_F360_Mod_divisor_with_decimals \(\)](#)
 - [f360_math_func_Test_F360_Mod_divisor_without_decimals \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix \(\)](#)
 - [f360_math_func_Test_F360_Multiply_2D_Matrix_With_Constant \(\)](#)
 - [f360_math_func_Test_F360_Translate_2D_Position \(\)](#)

2.23.1 Detailed Description

2.23.2 Function Documentation

2.23.2.1 f360_math_func()

test_group f360_math_func ()

Tests for f360_math_func.

Setup:

Setting up float32_t comparison thresholds for passing tests

Teardown:

Nothing to teardown in this test group

2.23.2.2 f360_math_func_F360_Linear_Equation()

f360_math_func_F360_Linear_Equation ()

Purpose: Check if function returns the correct value.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Verify result for lines when slope is zero.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.3 f360_math_func_F360_Linear_Equation_small_den()

```
f360_math_func_F360_Linear_Equation_small_den ( )
```

Purpose: Check if function behaves correctly for small denominators.

Requirements NA

Step 1:

Test behavior of zero and close to zero denominator else - first if $((x < x1) \ \&\& \ (x1 < x2))$

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Test behavior of zero and close to zero denominator else - first if $((x > x1) \ \&\& \ (x1 > x2))$

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 3:

Test behavior of zero and close to zero denominator else - second if $((x < x2) \ \&\& \ (x2 < x1))$

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Test behavior of zero and close to zero denominator else - second if ((x > x2) && (x2 > x1))

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Test behavior of zero and close to zero denominator else - else (x1 == x2)

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Test behavior of zero and close to zero denominator else - else (x1 == x2) with nonzero x1,x2 and x1>x2

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 7:\n

Test behavior of zero and close to zero denominator else - else ($x1 == x2$) with nonzero $x1, x2$ and $x2 > x1$

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.4 f360_math_func_F360_Linear_Equation_With_Saturation_small_den()

```
f360_math_func_F360_Linear_Equation_With_Saturation_small_den ( )
```

Purpose:\n Check if F360_Linear_Equation_With_Saturation behaves correctly for infinite slopes.

Requirements NA

Step 1:\n

Compare function output to expected output when not saturating and slope of line is close to infinite.

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 2:\n

Compare function output to expected output when saturating due to too large x query value and slope of line is close to infinite.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 3:\n

Compare function output to expected output when saturating due to too small x query value and slope of line is close to infinite.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Compare function output to expected output when not saturating and slope of line is infinite.

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Compare function output to expected output when saturating due to too large x query value and slope of line is infinite.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Compare function output to expected output when saturating due to too small x query value and slope of line is infinite.

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.5 f360_math_func_F360_Matmul_MxN_NxP()

f360_math_func_F360_Matmul_MxN_NxP ()

Purpose: Check if function returns the correct matrix.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Compare function output to expected output when the maximum matrix sizes are larger than the utilized matrix sizes.

Precondition:

Set up float32_t values to operate on and compute expected result

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.6 f360_math_func_F360_Matmul_MxN_NxP_Transpose()

```
f360_math_func_F360_Matmul_MxN_NxP_Transpose ( )
```

Purpose: Check if function returns the correct matrix.

Requirements NA

Step 1:

Compare function output to expected output using full matrices.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Compare function output to expected output using partially empty matrices.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.7 f360_math_func_F360_Saturate()

```
f360_math_func_F360_Saturate ( )
```

Purpose: Check if function returns the correct value.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.8 f360_math_func_F360_Sort_asc()

```
f360_math_func_F360_Sort_asc ( )
```

Purpose: Check if function returns the correct value.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.9 f360_math_func_F360_Sort_dsc()

```
f360_math_func_F360_Sort_dsc ( )
```

Purpose: Check if function returns the correct value.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.10 f360_math_func_F360_Sort_fail()

```
f360_math_func_F360_Sort_fail ( )
```

Purpose: Check that function is not sorting and return false when asked to sort more values than elements in array.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on. Expected values are identical as no change is expected.

Action:

Call function.

Result:

Ensure function output is equal to the expected result. No sorting of input value array and index array and return flag from sorting should be false.

2.23.2.11 f360_math_func_F360_Sort_no_sort()

```
f360_math_func_F360_Sort_no_sort ( )
```

Purpose: Check that sorting function will do no sorting when num_elements is zero.

Requirements NA

Precondition:

Set up array for sorting, and it's backup. Prepare expected permutation indexes after sorting.

Action:

Call function.

Result:

Ensure function output is equal to the expected result. Compare all indexes of input and its values. It should be same before and after the call to sorting function. Sorting function should return true.

2.23.2.12 f360_math_func_F360_Sort_partial()

```
f360_math_func_F360_Sort_partial ( )
```

Purpose: Check that sorting function can sort only specified portion of input. (Leaving remainder of input unsorted.)

Requirements NA

Precondition:

Set up array for sorting, and it's backup. Prepare expected permutation indexes after sorting.

Action:

Call function.

Result:

Ensure function output is equal to the expected result. Compare sorted(first 3) and also unsorted(last 2) indexes of inputs. All should match to expected idx_arr_asc. Observe that last two values remain unsorted. Only po

2.23.2.13 f360_math_func_F360_Sort_sorted_values_not_modified()

```
f360_math_func_F360_Sort_sorted_values_not_modified ( )
```

Purpose: Check that sorting function just sorts, but does not modify sorted values.

Requirements NA

Precondition:

Set up array for sorting, and it's backup. Prepare expected permutation indexes after sorting.

Action:

Call function.

Result:

Ensure function output is equal to the expected result. Permutation saved in idx_arr should map to original unsorted array. Floats must be equal without any tolerance, it is copy!

2.23.2.14 f360_math_func_F360_Sort_stability()

```
f360_math_func_F360_Sort_stability ( )
```

Purpose: Check that sorting is stable (preserving order of elements of same value).

Requirements NA

Precondition:

Set up array for sorting, and it's backup. Prepare expected permutation indexes after sorting.

Action:

Call function.

Result:

Ensure function output is equal to the expected result. Compare indexes of sorted array to expected indexes.

2.23.2.15 f360_math_func_F360_Sort_success()

```
f360_math_func_F360_Sort_success ( )
```

Purpose: Check that function returns true when sorting.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on.

Action:

Call function.

Result:

When sorting is done, sorting_success should return true.

2.23.2.16 f360_math_func_Test_F360_2d_Matrix_Determinant()

```
f360_math_func_Test_F360_2d_Matrix_Determinant ( )
```

Purpose: Check if function returns the correct determinant value.

Requirements NA

Step 1:

Test behaviour of function when inputting a matrix with all elements zero

Precondition:

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 2:\n

Test behaviour of function when inputting a matrix with all elements zero except 1 (case 1/4, first element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 3:\n

Test behaviour of function when inputting a matrix with all elements zero except 1 (case 2/4, second element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Test behaviour of function when inputting a matrix with all elements zero except 1 (case 3/4, third element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Test behaviour of function when inputting a matrix with all elements zero except 1 (case 4/4, fourth element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 1/12, first and second element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 7:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 2/12, first and third element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 8:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 3/12, second and fourth element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 9:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 4/12, third and fourth element is non-zero)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 10:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 5/12, first element positive fourth element positive)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 11:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 6/12, first element positive fourth element negative)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 12:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 7/12, first element negative fourth element positive)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 13:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 8/12, first element negative fourth element negative)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 14:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 9/12, second element positive third element positive)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 15:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 10/12, second element positive third element negative)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 16:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 11/12, second element negative third element positive)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 17:\n

Test behaviour of function when inputting a matrix with all elements zero except 2 (case 12/12, second element negative third element negative)

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 18:\n

Test behaviour of function when inputting a full positive definite covariance matrix

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 19:\n

Test behaviour of function when inputting a full arbitrary matrix

Precondition:\n

Set up matrix and expected output determinant value.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.17 f360_math_func_Test_F360_Accumulate_Probabilities()

```
f360_math_func_Test_F360_Accumulate_Probabilities ( )
```

Purpose: Check if F360_Accumulate_Probabilities returns correct output.

Requirements NA

Step 1:

Compare function output to expected output for input probabilities in the range of [0,1].

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.18 f360_math_func_test_f360_get_hypotenuse_negative_input()

```
f360_math_func_test_f360_get_hypotenuse_negative_input ( )
```

Purpose: Check if F360_Get_Hypotenuse returns the correct value when inputs are negative.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.19 f360_math_func_test_f360_get_hypotenuse_positive_input()

```
f360_math_func_test_f360_get_hypotenuse_positive_input ( )
```

Purpose:\n Check if F360_Get_Hypotenuse returns the correct value when inputs are positive.

Requirements NA

Step 1:\n

Compare function output to expected output.

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.20 f360_math_func_test_f360_get_hypotenuse_squared_negative_input()

```
f360_math_func_test_f360_get_hypotenuse_squared_negative_input ( )
```

Purpose:\n Check if F360_Get_Hypotenuse_Squared returns the correct value when inputs are negative.

Requirements NA

Step 1:\n

Compare function output to expected output.

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.21 f360_math_func_test_f360_get_hypotenuse_squared_positive_input()

```
f360_math_func_test_f360_get_hypotenuse_squared_positive_input ( )
```

Purpose:\n Check if F360_Get_Hypotenuse_Squared returns the correct value when inputs are positive.

Requirements NA

Step 1:\n

Compare function output to expected output.

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.22 f360_math_func_test_f360_get_hypotenuse_squared_zero_input()

```
f360_math_func_test_f360_get_hypotenuse_squared_zero_input ( )
```

Purpose:\n Check if F360_Get_Hypotenuse_Squared returns the correct value when inputs are zero.

Requirements NA

Step 1:\n

Compare function output to expected output.

Precondition:\n

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.23 f360_math_func_test_f360_get_hypotenuse_zero_input()

```
f360_math_func_test_f360_get_hypotenuse_zero_input ( )
```

Purpose: Check if F360_Get_Hypotenuse returns the correct value when inputs are zero.

Requirements NA

Step 1:

Compare function output to expected output.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.24 f360_math_func_Test_F360_Linear_Equation_With_Saturation()

```
f360_math_func_Test_F360_Linear_Equation_With_Saturation ( )
```

Purpose: Check if F360_Linear_Equation_With_Saturation returns the correct value for finite slopes.

Requirements NA

Step 1:

Compare function output to expected output when not saturating and slope of line is non-zero and finite.

Precondition:

Set up float32_t values to operate on and compute expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 2:\n

Compare function output to expected output when saturating due to too large x query value and slope of line is non-zero and finite.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 3:\n

Compare function output to expected output when saturating due to too small x query value and slope of line is non-zero and finite.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Compare function output to expected output when not saturating and slope of line is zero.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Compare function output to expected output when saturating due to too large x query value and slope of line is zero.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Compare function output to expected output when saturating due to too large x query value and slope of line is zero.

Precondition:\n

Set up expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.25 f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order()

```
f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order ( )
```

Purpose:\n Check if F360_Low_Pass_Filter_Angle_First_Order returns correct output for values far from edges $\pm\pi$.

Requirements NA

Step 1:\n

Compare function output to expected low pass filtered value

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.26 f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas()

```
f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas ( )
```

Purpose:\n Check if F360_Low_Pass_Filter_Angle_First_Order returns correct output where old value $> -\pi$ and measured value $< \pi$.

Requirements NA

Step 1:\n

Compare function output to expected low pass filtered value

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.27 f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas()

```
f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas ( )
```

Purpose: Check if F360_Low_Pass_Filter_Angle_First_Order returns correct output where old value $< \pi$ and measured value $> -\pi$.

Requirements NA

Step 1:

Compare function output to expected low pass filtered value

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.28 f360_math_func_Test_F360_Low_Pass_Filter_First_Order()

```
f360_math_func_Test_F360_Low_Pass_Filter_First_Order ( )
```

Purpose: Check if F360_Low_Pass_Filter_First_Order returns correct output.

Requirements NA

Step 1:

Compare function output to expected low pass filtered value

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.29 f360_math_func_Test_F360_matmul_2x2_2x1()

```
f360_math_func_Test_F360_matmul_2x2_2x1 ( )
```

Purpose:\n Check if F360_matmul_2x2_2x1 returns correct 2-by-2 and 2-by-1 matrix multiplication

Requirements NA

Step 1:\n

Check that correct output is returned.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.30 f360_math_func_Test_F360_Mod_divisor_with_decimals()

```
f360_math_func_Test_F360_Mod_divisor_with_decimals ( )
```

Purpose:\n Check if F360_Mod returns correct output when the divisor is a float32_t with decimals

Requirements NA

Step 1:\n

Check that correct output is returned when both values are positive.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 2:\n

Check that correct output is returned when dividend is negative and divisor is positive.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 3:\n

Check that correct output is returned when dividend is positive and divisor is negative.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Check that correct output is returned when dividend is positive and divisor is negative.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Check that correct output is returned when divisor is close to zero and negative.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Check that correct output is returned when divisor is close to zero and positive.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.31 f360_math_func_Test_F360_Mod_divisor_without_decimals()

```
f360_math_func_Test_F360_Mod_divisor_without_decimals ( )
```

Purpose: Check if F360_Mod returns correct output when the divisor is a truncated float32_t value without decimals.

Requirements NA

Step 1:

Check that correct output is returned when both values are positive.

Precondition:

Set up values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Check that correct output is returned when dividend is negative and divisor is positive.

Precondition:

Set up values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 3:

Check that correct output is returned when dividend is positive and divisor is negative.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Check that correct output is returned when both input arguments are negative.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Check that correct output is returned when divider is zero.

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.32 f360_math_func_Test_F360_Multiply_2D_Matrix_With_Constant()

```
f360_math_func_Test_F360_Multiply_2D_Matrix_With_Constant ( )
```

Purpose: Check that function F360_Multiply_2D_Matrix_With_Constant is

correctly multiplying each element of a matrix with a constant.

Requirements NA

Step 1:

Check that output matrix is a zero matrix when input matrix is a zero matrix

Precondition:

Set up a zero input matrix, a non-zero constant and an expected output zero matrix

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Check that output matrix is a zero matrix when input constant is zero

Precondition:

Set up a non-zero input matrix, a zero constant and an expected output zero matrix

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 3:

Check that output matrix is correct when input matrix is non-zero and input constant is positive

Precondition:\n

Set up a non-zero input matrix, a positive constant and an expected output matrix

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Check that output matrix is correct when input matrix is non-zero and input constant is negative

Precondition:\n

Set up a non-zero input matrix, a negative constant and an expected output matrix

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Check that output matrix is correct when input matrix has more rows and columns than is actually used

Precondition:\n

Set up a non-zero input matrix with too many rows and columns (2x3 matrix is used but number of rows and columns are 20x13), a positive constant and an expected output matrix

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.33 f360_math_func_Test_F360_Power_Scale_Prob()

```
f360_math_func_Test_F360_Power_Scale_Prob ( )
```

Purpose: Check if F360_F360_Power_Scale_Prob returns correct output.

Requirements NA

Step 1:

Compare function output to expected output for exponent within the range [1, 25] and probability within the range [0,1].

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Compare function output to expected output for exponent within the range [1, 25] and probability below 0.

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 3:

Compare function output to expected output for exponent within the range [1, 25] and probability above 1.

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Compare function output to expected output for exponent below 1 and probability in the range [0,1].

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Compare function output to expected output for exponent above 25 and probability in the range [0,1].

Precondition:\n

Set up float32_t values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.34 f360_math_func_Test_F360_Safe_Logf()

```
f360_math_func_Test_F360_Safe_Logf ( )
```

Purpose: Check if F360_Safe_Logf returns correct output.

Requirements NA

Step 1:

Compare function output to expected output for non-zero input.

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

Step 2:

Compare function output to expected output for zero input.

Precondition:

Set up float32_t values to operate on and expected result.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.35 f360_math_func_Test_F360_Translate_2D_Position()

```
f360_math_func_Test_F360_Translate_2D_Position ( )
```

Purpose: Check that function F360_Translate_2D_Position works as intended.

Requirements NA

Precondition:

Set up point to be translated and translation vector properties.

Action:

Call function.

Result:

Ensure function output is equal to the expected result.

2.23.2.36 f360_math_func_Test_Rotate_2D_Covariance_Matrix()

```
f360_math_func_Test_Rotate_2D_Covariance_Matrix ( )
```

Purpose: Check if function Rotate_2D_Covariance_Matrix is correctly

transforming a covariance matrix between two differently oriented coordinate systems. Testing rotation angles from all 4 different quadrants.

Requirements NA

Step 1:

Check that covariance matrix is unchanged when rotating pi radians

Precondition:

Set up values to operate on and expected result.

Action:

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 2:\n

Check that covariance matrix is unchanged when rotating $-\pi$ radians

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 3:\n

Check that rotated covariance matrix is correct when rotating with an angle from fourth quadrant

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 4:\n

Check that rotated covariance matrix is correct when rotating with an angle from third quadrant

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 5:\n

Check that rotated covariance matrix is correct when rotating with an angle from second quadrant

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

Step 6:\n

Check that rotated covariance matrix is correct when rotating with an angle from first quadrant

Precondition:\n

Set up values to operate on and expected result.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.23.2.37 f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5()

f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5 ()

Purpose:\n Check if function Rotate_2D_Covariance_Matrix_With_Precalc_Coeff

correctly returns values of Covariance coefficients for $6\pi/5$ angle.

Requirements NA

Precondition:\n

Set up values to operate on and expected result for $6\pi/5$ angle.

Action:\n

Call function Rotate_2D_Covariance_Matrix_With_Precalc_Coeff()

Result:\n

Ensure function output is equal to the expected result.

2.23.2.38 f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3()

```
f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3 ( )
```

Purpose: Check if function `Rotate_2D_Covariance_Matrix_With_Precalc_Coeff` correctly returns values of Covariance coefficients for $-\pi/3$ angle.

Requirements NA

Precondition:

Set up values to operate on and expected result for $-\pi/3$ angle.

Action:

Call function `Calculate_Rotated_2D_Covariance_Coefficients()`

Result:

Ensure function output is equal to the expected result.

2.23.2.39 f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle()

```
f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle ( )
```

Purpose: Check if function `Rotate_2D_Covariance_Matrix_With_Precalc_Coeff` correctly returns values of Covariance coefficients for π angle.

Requirements NA

Precondition:

Set up values to operate on and expected result for π angle

Action:

Call function `Rotate_2D_Covariance_Matrix_With_Precalc_Coeff()`

Result:

Ensure function output is equal to the expected result.

2.23.2.40 f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3()

```
f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3 ( )
```

Purpose:\n Check if function Calculate_Rotated_2D_Covariance_Coefficients
correctly returns values of Covariance coefficients for $\pi/3$ angle.

Requirements NA

Precondition:\n
Set up values to operate on and expected result for $\pi/3$ angle.

Action:\n
Call function Rotate_2D_Covariance_Matrix_With_Precalc_Coeff()

Result:\n
Ensure function output is equal to the expected result.

2.23.2.41 f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints()

```
f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints ( )
```

Purpose:\n Check that RSPP_Piecewise_Linear_Equation returns the correct
value when the input breakpoints are very close.

Requirements NA

Step 1:\n
Compare function output to expected output.

Precondition:\n
Set up breakpoints, query points and expected result.

Action:\n
Call function with query point 1.

Result:\n
Ensure function output is equal to the expected result.

2.23.2.42 f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints()

```
f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints ( )
```

Purpose:\n Check that RSPP_Piecewise_Linear_Equation returns the correct value when the input breakpoints are sorted.

Requirements NA

Step 1:\n

Compare function output to expected output.

Precondition:\n

Set up breakpoints, query points and expected result.

Action:\n

Call function with query point 1.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 2.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 3.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 4.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 5.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 6.

Result:\n

Ensure function output is equal to the expected result.

Action:\n

Call function with query point 7.

Result:\n

Ensure function output is equal to the expected result.

2.24 F360_math_func_f360_rotate_2d_vector

Functions

- test group [f360_math_func_f360_rotate_2d_vector \(\)](#)

Tests for f360_math_func_f360_rotate_2d_vector.

- [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Zero \(\)](#)
- [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_180 \(\)](#)
- [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_90 \(\)](#)
- [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90 \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_1 \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_2 \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Slightly_Smaller_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Slightly_Larger_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Greater_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Equal_To_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Residual_Value_Slightly_Smaller_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Slightly_Greater_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Correct_Result_For_Valid_Input \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Negative_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Smaller_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Greater_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Equal_To_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Absolute_Value_Of_Negative_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Absolute_Value_Of_Negative_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Both_Input_Values_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Residual_Equal_To_Zero_And_Non_Zero_Tuning_Constant \(\)](#)
- [f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_No_Slope \(\)](#)
- [f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_Non_Zero_Slope \(\)](#)
- [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_No_Slope \(\)](#)
- [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_Non_Zero_Slope \(\)](#)
- [f360_math_func_Test_F360_Cross_Product_Of_Vectors \(\)](#)
- [f360_math_func_Test_F360_Cross_Product_Of_Negative_Vectors \(\)](#)

- [f360_math_func_Test_F360_Cross_Product_Of_Parallel_Vectors \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Positive_K_Zero_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Negative_K_Positive_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Negative_Infinite_K_Zero_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Positive_Infinite_K_Positive_M \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Two_Lines_Parallel_Lines \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Two_Lines_Intersection_Exists \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Imaginary_Root \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Negative_A_Coefficient \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Positive_A_Coefficient \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Small_Slope \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Large_Slope \(\)](#)

2.24.1 Detailed Description

2.24.2 Function Documentation

2.24.2.1 f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_No_Slope()

`f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_No_Slope ( )`

Purpose:\n

Test that squared distance from a point to a line without any slope is calculated correctly

Requirements NA

Precondition:\n

Define a point to evaluate and create a line without any slope

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.2 f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_Non_Zero_Slope()

```
f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_Non_Zero_Slope ( )
```

Purpose:\n

Test that squared distance from point to a line with non-zero slope is calculated correctly

Requirements NA

Precondition:\n

Define a point to evaluate and create a line with non-zero slope

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.3 f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_No_Slope()

```
f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_No_Slope ( )
```

Purpose:\n

Test that projection of point on a line without any slope is calculated correctly

Requirements NA

Precondition:\n

Define a point to evaluate and create a line without a slope

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.4 f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_Non_Zero_Slope()

f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_Non_Zero_Slope ()

Purpose:\n

Test that projection of point on a line with non-zero slope is calculated correctly

Requirements NA

Precondition:\n

Define a point to evaluate and create a line with non-zero slope

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.5 f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_1()

f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_1 ()

Purpose:\n

Function should return a correct result for example input values

Requirements NA

Precondition:\n

Set example input values

Action:\n

Call tested function

Result:\n

Check expected result for example input values

2.24.2.6 f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_2()

f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_2 ()

Purpose:\n

Function should return a correct result for example input values

Requirements NA

Precondition:\n

Set example input values

Action:\n

Call tested function

Result:\n

Check expected result for example input values

2.24.2.7 f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Residual_Value_Slightly_Smaller_Than_Tuning_Constant()

f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Residual_Value_Slightly_Smaller_Than_Tuning_Constant ()

Purpose:\n

Function should return valid result for residual slightly smaller than tuning_constant

Requirements NA

Precondition:\n

Set input values according to test purpose

Action:\n

Call tested function

Result:\n

Check expected result for example input values

2.24.2.8 f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Slightly_Larger_Than_Min_Denominator ()

```
f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Slightly_↵  
Larger_Than_Min_Denominator ( )
```

Purpose:\n

Function should return valid result for tuning_constant value slightly larger than EPSILON

Requirements NA

Precondition:\n

Set input values according to test purpose

Action:\n

Call tested function

Result:\n

Check expected result for given input values

2.24.2.9 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Equal_To_Tuning_Constant()

```
f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Equal_To_Tuning_↵  
Constant ( )
```

Purpose:\n

Function should return zero for residual equal to tuning_constant

Requirements NA

Precondition:\n

Set input values according to test purpose

Action:\n

Call tested function

Result:\n

Check expected result for example input values

2.24.2.10 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Greater_Than_Tuning_Constant

f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Greater_Than_Tuning_Constant ()

Purpose:

Function should return zero for x greater than tuning_constant

Requirements NA

Precondition:

Set input values according to test purpose

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.11 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Slightly_Greater_Than_Tuning_Constant

f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Slightly_Greater_Than_Tuning_Constant ()

Purpose:

Function should return zero for residual slightly greater than tuning_constant

Requirements NA

Precondition:

Set input values according to test purpose

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.12 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator

f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator ()

Purpose:

Function should return zero for tuning_constant value equal to EPSILON

Requirements NA

Precondition:

Set input values according to test purpose

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.13 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero()

f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero ()

Purpose:

Function should return zero for tuning_constant value equal to zero

Requirements NA

Precondition:

Set input values according to test purpose

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.14 f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Slightly_Smaller_Than_Min_Denominator

f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Slightly_Smaller_Than_Min_Denominator ()

Purpose:

Function should return zero for tuning_constant value slightly smaller than EPSILON

Requirements NA

Precondition:

Set input values according to test purpose

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.15 f360_math_func_F360_Huber_Weight_Should_Return_Correct_Result_For_Valid_Input()

f360_math_func_F360_Huber_Weight_Should_Return_Correct_Result_For_Valid_Input ()

Purpose:* Function should return a correct result for valid input values

Requirements NA

Precondition:

Set example input values

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.16 f360_math_func_F360_Huber_Weight_Should_Return_One_For_Residual_Equal_To_Zero_And_Non_Zero_Tuning_Constant

```
f360_math_func_F360_Huber_Weight_Should_Return_One_For_Residual_Equal_To_Zero_And_Non_Zero_Tuning_Constant ( )
```

Purpose:

Function should return one for both residual equal to zero and non-zero tuning_constant

Requirements NA

Precondition:

Set residual equal to zero and tuning_constant to non_zero value

Action:

Call tested function

Result:

Check for valid output value

2.24.2.17 f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Equal_To_Positive_Residual

```
f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Equal_To_Positive_Residual ( )
```

Purpose:

Function should return one for tuning_constant equal to positive residual

Requirements NA

Precondition:

Set tuning_constant equal to positive residual

Action:

Call tested function

Result:

Check expected result for example input values

2.24.2.18 f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Absolute_Valu

```
f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Absolute↵  
_Value_Of_Negative_Residual ( )
```

Purpose:

Function should return one for tuning_constant greater than positive residual

Requirements NA

Precondition:

Set tuning_constant to value greater than absolute value of negative residual

Action:

Call tested function

Result:

Function result should be equal to one

2.24.2.19 f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Positive_Resid

```
f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Positive↵  
_Residual ( )
```

Purpose:

Function should return one for tuning_constant greater than positive residual

Requirements NA

Precondition:

Set tuning_constant to value greater than positive residual

Action:

Call tested function

Result:

Function result should be equal to one

2.24.2.20 f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Absol

```
f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than↵_Absolute_Value_Of_Negative_Residual ( )
```

Purpose:\n

Function should return valid result for tuning_constant smaller than absolute value of negative residual

Requirements NA

Precondition:\n

Set tuning_constant smaller than absolute value of negative residual

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.21 f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Positiv

```
f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than↵_Positive_Residual ( )
```

Purpose:\n

Function should return valid result for tuning_constant smaller than positive residual

Requirements NA

Precondition:\n

Set tuning_constant smaller than positive residual

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.22 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Both_Input_Values_Equal_To_Zero()

f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Both_Input_Values_Equal_To_Zero ()

Purpose:\n

Function should return zero for both input parameters equal to zero

Requirements NA

Precondition:\n

Set both input value to zero

Action:\n

Call tested function

Result:\n

Check for valid output value

2.24.2.23 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Negative_Tuning_Constant()

f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Negative_Tuning_Constant ()

Purpose:\n

Function should return zero for negative value of tuning_constant

Requirements NA

Precondition:\n

Set tuning_constant to negative value

Action:\n

Call tested function

Result:\n

Function result should be equal to zero

2.24.2.24 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator()

```
f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator  
( )
```

Purpose:

Function should return zero for tuning_constant value equal to EPSILON

Requirements NA

Precondition:

Set tuning_constant equal to EPSILON

Action:

Call tested function

Result:

Function result should be equal to zero

2.24.2.25 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero()

```
f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero ( )
```

Purpose:

Function should return zero for tuning_constant equal to zero

Requirements NA

Precondition:

Set tuning_constant equal to zero

Action:

Call tested function

Result:

Function result should be equal to zero

2.24.2.26 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Greater_Than_Min_Denominator

f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Greater_Than_Min_Denominator ()

Purpose:

Function should return valid result for tuning_constant value slightly greater than EPSILON

Requirements NA

Precondition:

Set tuning_constant equal slightly greater than EPSILON

Action:

Call tested function

Result:

Function result return a valid result

2.24.2.27 f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Smaller_Than_Min_Denominator

f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Smaller_Than_Min_Denominator ()

Purpose:

Function should return zero for tuning_constant value slightly smaller than EPSILON

Requirements NA

Precondition:

Set tuning_constant equal slightly smaller than EPSILON

Action:

Call tested function

Result:

Function result should be equal to zero

2.24.2.28 f360_math_func_f360_rotate_2d_vector()

```
test_group f360_math_func_f360_rotate_2d_vector ( )
```

Tests for f360_math_func_f360_rotate_2d_vector.

Setup:\n

Set up float32_t comparison threshold for passing tests Set a fixed initial position Declare common variables used within all tests

Teardown:\n

Nothing to teardown in this test group

2.24.2.29 f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_180()

```
f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_180 ( )
```

Purpose:\n Check that function F360_Rotate_2D_Vector works as intended when angle is 180 degrees.

Requirements NA

Precondition:\n

Set up vector to be rotated and rotation angle properties when angle is 180 degrees.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.24.2.30 f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_90()

```
f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_90 ( )
```

Purpose:\n Check that function F360_Rotate_2D_Vector works as intended when angle is 90 degrees.

Requirements NA

Precondition:\n

Set up vector to be rotated and rotation angle properties when angle is 90 degrees.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.24.2.31 f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90()

```
f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90 ( )
```

Purpose:\n Check that function F360_Rotate_2D_Vector works as intended when angle is -90 degrees.

Requirements NA

Precondition:\n

Set up vector to be rotated and rotation angle properties when angle is -90 degrees.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.24.2.32 f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Zero()

```
f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Zero ( )
```

Purpose:\n Check that function F360_Rotate_2D_Vector works as intended when angle is zero.

Requirements NA

Precondition:\n

Set up vector to be rotated and rotation angle properties when angle is 0 degrees.

Action:\n

Call function.

Result:\n

Ensure function output is equal to the expected result.

2.24.2.33 f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Large_Slope

```
f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Large←  
_Slope ( )
```

Purpose:\n

Test finding of intersection between a line and 2 degree polynomial works correctly when the a-coefficient is -1, b-coefficient is 0 and c-coefficient is 1. The line has a large slope of 1000 meaning that we expect one of the intersections close to zero.

Requirements NA

Precondition:\n

Define a polynomial and a line Initialize resulting x intersection variables

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.34 f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Small_Slope

```
f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Small_Slope ( )
```

Purpose:

Test finding of intersection between a line and 2 degree polynomial works correctly when the a-coefficient is -1, b-coefficient is 0 and c-coefficient is 1. The line has a slope of 1.

Requirements NA

Precondition:

Define a polynomial and a line Initialize resulting x intersection variables

Action:

Call function

Result:

Check if result match to expected value.

2.24.2.35 f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Imaginary_Root()

```
f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Imaginary_Root ( )
```

Purpose:

Test finding of intersection between a line and 2 degree polynomial works correctly when the intersection is imaginary.

Requirements NA

Precondition:

Define a polynomial and a line Initialize resulting x intersection variables

Action:

Call function

Result:

Check if result match to expected value.

2.24.2.36 f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Negative_A_Coefficient()

```
f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Negative_A_Coefficient ( )
```

Purpose:

Test finding of intersection between a line and 2 degree polynomial works correctly when the a-coefficient is very small and negative. When a-coefficient is very small we get floating point precision errors and even though we expect $x = 1$ as a solution we actually get 0.

Requirements NA**Precondition:**

Define a polynomial and a line Initialize resulting x intersection variables

Action:

Call function

Result:

Check if result match to expected value.

2.24.2.37 f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Positive_A_Coefficient()

```
f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Positive_A_Coefficient ( )
```

Purpose:

Test finding of intersection between a line and 2 degree polynomial works correctly when the a-coefficient is very small and negative. When a-coefficient is very small we get floating point precision errors and even though we expect $x = 1$ as a solution we actually get 0.

Requirements NA**Precondition:**

Define a polynomial and a line Initialize resulting x intersection variables

Action:

Call function

Result:

Check if result match to expected value.

2.24.2.38 f360_math_func_Find_X_Intersect_Between_Two_Lines_Intersection_Exists()

```
f360_math_func_Find_X_Intersect_Between_Two_Lines_Intersection_Exists ( )
```

Purpose:\n

Test finding of intersection between two lines works as expected when an intersection exists

Requirements NA

Precondition:\n

Define two lines with different slopes and offset Initialize resulting x intersection variable

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.39 f360_math_func_Find_X_Intersect_Between_Two_Lines_Parallel_Lines()

```
f360_math_func_Find_X_Intersect_Between_Two_Lines_Parallel_Lines ( )
```

Purpose:\n

Test finding of intersection between two lines works as expected when the lines are parallel

Requirements NA

Precondition:\n

Define two parallel lines Initialize resulting x intersection variable

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.40 f360_math_func_Get_Vector_As_Linear_Equation_Negative_Infinite_K_Zero_M()

```
f360_math_func_Get_Vector_As_Linear_Equation_Negative_Infinite_K_Zero_M ( )
```

Purpose:\n

Test that conversion from finite vector to linear equation works as expected when x start and end points are almost equal.

Requirements NA

Precondition:\n

Define a finite vector Initialize resulting k and m variables

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.41 f360_math_func_Get_Vector_As_Linear_Equation_Negative_K_Positive_M()

```
f360_math_func_Get_Vector_As_Linear_Equation_Negative_K_Positive_M ( )
```

Purpose:\n

Test that conversion from finite vector to linear equation works as expected when we expect a negative slope with positive offset.

Requirements NA

Precondition:\n

Define a finite vector Initialize resulting k and m variables

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.42 f360_math_func_Get_Vector_As_Linear_Equation_Positive_Infinite_K_Positive_M()

```
f360_math_func_Get_Vector_As_Linear_Equation_Positive_Infinite_K_Positive_M ( )
```

Purpose:\n

Test that conversion from finite vector to linear equation works as expected when x start and end points are almost equal.

Requirements NA

Precondition:\n

Define a finite vector Initialize resulting k and m variables

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.43 f360_math_func_Get_Vector_As_Linear_Equation_Positive_K_Zero_M()

```
f360_math_func_Get_Vector_As_Linear_Equation_Positive_K_Zero_M ( )
```

Purpose:\n

Test that conversion from finite vector to linear equation works as expected when we expect a positive slope with no offset.

Requirements NA

Precondition:\n

Define a finite vector Initialize resulting k and m variables

Action:\n

Call function

Result:\n

Check if result match to expected value.

2.24.2.44 f360_math_func_Test_F360_Cross_Product_Of_Negative_Vectors()

```
f360_math_func_Test_F360_Cross_Product_Of_Negative_Vectors ( )
```

Purpose:\n

Test that Cross Product of two 2-D vectors is calculated correctly.

Requirements NA

Precondition:\n

Define vectors' parameters.

Action:\n

Call F360_Cross_Product

Result:\n

Check if result match to expected value.

2.24.2.45 f360_math_func_Test_F360_Cross_Product_Of_Parallel_Vectors()

```
f360_math_func_Test_F360_Cross_Product_Of_Parallel_Vectors ( )
```

Purpose:\n

Test that Cross Product of two 2-D vectors is calculated correctly. Vectors are parallel to each other.

Requirements NA

Precondition:\n

Define parameters for parallel vectors.

Action:\n

Call F360_Cross_Product

Result:\n

Check if result match to expected value.

2.24.2.46 f360_math_func_Test_F360_Cross_Product_Of_Vectors()

f360_math_func_Test_F360_Cross_Product_Of_Vectors ()

Purpose:

Test that Cross Product of two 2-D vectors is calculated correctly.

Requirements NA

Precondition:

Define vectors' parameters.

Action:

Call F360_Cross_Product

Result:

Check if result match to expected value.

2.25 Clamp

Functions

- test group `clamp()`

Test group of Clamp function. Tests verify whether function properly clamps variable into provided limits.

2.25.1 Detailed Description

2.26 F360_norm_heading_angle

Functions

- test group `f360_norm_heading_angle` ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- `f360_norm_heading_angle_FunctionToTest_Descriptive_Tag` ()

2.26.1 Detailed Description

2.26.2 Function Documentation

2.26.2.1 f360_norm_heading_angle()

```
test group f360_norm_heading_angle ( )
```

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

Teardown:\n

Describe what is done in test teardown. Remove test teardown function and this tag if it is not used.

2.26.2.2 f360_norm_heading_angle_FunctionToTest_Descriptive_Tag()

```
f360_norm_heading_angle_FunctionToTest_Descriptive_Tag ( )
```

Purpose:\n

Describe purpose of test on a high level. The description should be unique and distinguishable from other tests' purposes. E.g. test that function `Some_Function()` works as intended given some specific preconditions.

Requirements Put in requirement tag if any otherwise set to NA.

Precondition:\n

Describe prepared input. E.g. describe relevant set up from test group and what variables are changed or added for this specific test.

Action:\n

Describe the action of the test. E.g. call `Some_Function()`.

Result:\n

Describe expected output. E.g. check that the output match expected data.

2.27 F360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate ↩

Functions

- test group [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate](#) ()
This test group tests the functionality of the function `Get_Uncertainty_Of_Compensated_Range_Rate()`
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Inside_Range](#) ()
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_On_Boundary_Of_Range](#) ()
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Out_Of_Range](#) ()

2.27.1 Detailed Description

2.27.2 Function Documentation

2.27.2.1 f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Inside_Range()

```
f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Inside_Range
( )
```

Purpose:\n Test `Get_Uncertainty_Of_Compensated_Range_Rate` Function when both cos and sin of azimuth are inside allowed range

Requirements NA

Precondition:\n

Setting up `cos_det_az` to 0.5 and `sin_det_az` to 0.8

Action:\n

Call `Get_Uncertainty_Of_Compensated_Range_Rate` Function

Result:\n

Check the expected value for `var_comp_rng_rate`

2.27.2.2 f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_On_Boundary_Of_Ra

```
f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_On_Boundary↵  
_Of_Range ( )
```

Purpose:\n Test Get_Uncertainty_Of_Compensated_Range_Rate Function when cos or sin of azimuth is on boundary of allowed range

Requirements NA

Precondition:\n

Setting up cos_det_az to 1 and sin_det_az to 0

Action:\n

Call Get_Uncertainty_Of_Compensated_Range_Rate Function

Result:\n

Check the expected value for var_comp_rng_rate

2.27.2.3 f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Out_Of_Range()

```
f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Out_Of_Range  
( )
```

Purpose:\n Test Get_Uncertainty_Of_Compensated_Range_Rate Function when cos and sin of azimuth is beyond allowed range

Requirements NA

Precondition:\n

Setting up input to function and expected output

Action:\n

Call Get_Uncertainty_Of_Compensated_Range_Rate Function

Result:\n

Check the expected value for var_comp_rng_rate

2.28 F360_sensor_capability_Compute_Raw_Detection_Uncertainty

Functions

- test group [f360_sensor_capability_Compute_Raw_Detection_Uncertainty](#) ()
This test group tests the functionality of the function `Compute_Raw_Detection_Uncertainty()`
- [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty](#) ()
- [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty_2](#) ()

2.28.1 Detailed Description

2.28.2 Function Documentation

2.28.2.1 [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty\(\)](#)

```
f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty
( )
```

Purpose:\n Test `Compute_Raw_Detection_Uncertainty` Function with MRR360 radar type

Requirements NA

Precondition:\n

Setting up input to function and expected output

Action:\n

Call `Compute_Raw_Detection_Uncertainty` Function

Result:\n

Check the expected value for `rng_var`, `az_var` and `rng_rate_var`

2.28.2.2 f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty_2()

f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty_2 ()

Purpose: Test Compute_Raw_Detection_Uncertainty Function with MRR3 radar type

Requirements NA

Precondition:

Setting up input to function and expected output

Action:

Call Compute_Raw_Detection_Uncertainty Function

Result:

Check the expected value for rng_var, az_var and rng_rate_var

2.29 F360_sensor_capability_Get_Host_Velocity_Uncertainty

Functions

- test group [f360_sensor_capability_Get_Host_Velocity_Uncertainty](#) ()
This test group tests the functionality of the function [Get_Host_Velocity_Uncertainty](#)()
- [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_Forward_speed](#) ()
- [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_backward_speed](#) ()

2.29.1 Detailed Description

2.29.2 Function Documentation

2.29.2.1 f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_backward_speed()

`f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_backward_speed ( )`

Purpose:\n Test [Get_Host_Velocity_Uncertainty](#) function

Requirements NA

Precondition:\n

Setting up the Host speed to negative value

Action:\n

Call [Get_Host_Velocity_Uncertainty](#) function

Result:\n

Check the expected value for `velocity_cov`

2.29.2.2 f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_Forward_speed()

`f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_Forward_speed ( )`

Purpose:\n Test [Get_Host_Velocity_Uncertainty](#) function

Requirements NA

Precondition:\n

Setting up input to function and expected output

Action:\n

Call [Get_Host_Velocity_Uncertainty](#) function

Result:\n

Check the expected value for `velocity_cov`

2.30 F360_sensor_capability_Sensor_Capability_Detections

Functions

- test group [f360_sensor_capability_Sensor_Capability_Detections](#) ()
This test group tests the functionality of the function [Sensor_Capability_Detections\(\)](#)
- [f360_sensor_capability_Sensor_Capability_Detections_Test_Sensor_Capability_Detections](#) ()

2.30.1 Detailed Description

2.30.2 Function Documentation

2.30.2.1 f360_sensor_capability_Sensor_Capability_Detections_Test_Sensor_Capability_Detections()

`f360_sensor_capability_Sensor_Capability_Detections_Test_Sensor_Capability_Detections ( )`

Purpose:\n Test [Sensor_Capability_Detections](#) function

[Requirements](#) NA

Precondition:\n

Setting up input to function and expected output

Action:\n

Call [Sensor_Capability_Detections](#) function

Result:\n

Check the expected value for `std_vcs_az_scm`, `std_range_rate_compensated_scm`, `vcs_position_cov_scm` and `vcs_cross_covariances_scm`

2.31 F360_sensor_capability_Sensor_Capability_Host

Functions

- test group `f360_sensor_capability_Sensor_Capability_Host` ()
This test group tests the functionality of the function `Sensor_Capability_Host()`
- `f360_sensor_capability_Sensor_Capability_Host_Test_Sensor_Capability_Host` ()

2.31.1 Detailed Description

2.31.2 Function Documentation

2.31.2.1 `f360_sensor_capability_Sensor_Capability_Host_Test_Sensor_Capability_Host()`

`f360_sensor_capability_Sensor_Capability_Host_Test_Sensor_Capability_Host` ()

Purpose:\n Test `Sensor_Capability_Host` Function

Requirements NA

Precondition:\n

Setting up input to function and expected output

Action:\n

Call `Sensor_Capability_Host` function

Result:\n

Check the expected value for `position_inc_cov_scm`, `vel_cov_scm`, `std_speed_scm` and `std_yaw_rate_scm`

2.32 F360_sensor_capability

Functions

- test group [f360_sensor_capability](#) ()
This test group tests sensor_capability file.
- [f360_sensor_capability_Test_Compute_Raw_Host_Speed_Uncertainty](#) ()
- [f360_sensor_capability_Get_Uncertainty_Of_Detection_Position](#) ()
- [f360_sensor_capability_Test_Compute_Detection_Cross_Covariances](#) ()

2.32.1 Detailed Description

2.32.2 Function Documentation

2.32.2.1 f360_sensor_capability_Get_Uncertainty_Of_Detection_Position()

`f360_sensor_capability_Get_Uncertainty_Of_Detection_Position ( )`

Purpose:\n Test that detection range and azimuth uncertainty is correctly propagated to uncertainty in detection vcs position.

Requirements NA

Step 1:\n

Testing that Get_Uncertainty_Of_Detection_Position outputs the correct value

Precondition:\n

Setting up input to function and expected output

Action:\n

Call the function for some different azimuths

Result:\n

Check that function has output the expected values

2.32.2.2 f360_sensor_capability_Test_Compute_Detection_Cross_Covariances()

f360_sensor_capability_Test_Compute_Detection_Cross_Covariances ()

Purpose:\n

Check if Compute_Detection_Cross_Covariances returns the expected cross variance element.

Requirements NA

Step 1:\n

Testing that Compute_Detection_Cross_Covariances returns the correct value

Precondition:\n

Setting up input to function and expected output

Action:\n

Call the function Compute_Detection_Cross_Covariances

Result:\n

Check that function has output the expected values

Precondition:\n

Setting up another host heading angle and sensor motion as input to function and expected output

Action:\n

Call the function Compute_Detection_Cross_Covariances

Result:\n

Check that function has output the expected values

Precondition:\n

Setting up an negative host heading angle as input to function and expected output

Action:\n

Call the function Compute_Detection_Cross_Covariances

Result:\n

Check that function has output the expected values

2.32.2.3 f360_sensor_capability_Test_Compute_Raw_Host_Speed_Uncertainty()

f360_sensor_capability_Test_Compute_Raw_Host_Speed_Uncertainty ()

Purpose:\n Describe purpose of test

Requirements NA

Step 1:\n

Testing that Compute_Raw_Host_Speed_Uncertainty() behaves as expected

Precondition:\n

Setting up input to function and expected output

Action:\n

Call the function

Result:\n

Check that output corresponds to following two criteria: 1) speed and yaw rate variance is always ≥ 0 2) speed and yaw rate variance is increasing (or same but not decreasing) with $\text{abs}(\text{speed})$ and $\text{abs}(\text{yaw rate})$

2.33 F360_set_default_flags_raw_detections

Functions

- test group [f360_set_default_flags_raw_detections](#) ()
- [f360_set_default_flags_raw_detections_set_default_values_for_raw_detections](#) ()

2.33.1 Detailed Description

2.33.2 Function Documentation

2.33.2.1 f360_set_default_flags_raw_detections()

```
test_group f360_set_default_flags_raw_detections ( )
```

Setup:\n

Nothing to set up.

2.33.2.2 f360_set_default_flags_raw_detections_set_default_values_for_raw_detections()

```
f360_set_default_flags_raw_detections_set_default_values_for_raw_detections ( )
```

Purpose:\n

Test function that sets default values for detection flags (currently only `f_ok_to_use`).

Action:\n

- call `Set_Default_Flags_Raw_Detections()`

Result:\n

For all valid detections `f_ok_to_use` flag should be true.

2.34 F360_statistical_tests

Functions

- test group [f360_statistical_tests](#) ()
Test group of f360_statistical_tests.cpp file content. Tests verify whether statistical tests are properly performed.
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_calculated](#) ()
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_saturated](#) ()
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__test_variable_greater_than_max_test_variable](#) ()
- [f360_statistical_tests_F360_Try_To_Reject_Test__properly_rejects_hypothesis](#) ()
- [f360_statistical_tests_F360_Try_To_Reject_Test__properly_does_not_reject_hypothesis](#) ()
- [f360_statistical_tests_Check_Stationary_Hypothesis__handles_std_equal_to_zero](#) ()
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_and_test_variable_above_thresholds](#) ()
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_above_threshold_test_variable_below_threshold](#) ()
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_below_threshold_test_variable_above_threshold](#) ()

2.34.1 Detailed Description

2.34.2 Function Documentation

2.34.2.1 [f360_statistical_tests_Check_Stationary_Hypothesis__handles_std_equal_to_zero](#)()

`f360_statistical_tests_Check_Stationary_Hypothesis__handles_std_equal_to_zero ( )`

Purpose:\n

Purpose of this test is to verify whether standard deviation value equal to zero is properly handled

Requirements NA.

Precondition:\n

Set up abs test val Set up test value std Set up test value threshold Set up sigma threshold

Action:\n

Call tested function

Result:\n

Check whether test_result is false

2.34.2.2 f360_statistical_tests_Check_Stationary_Hypothesis__sigma_above_threshold_test_variable_below_threshold()

```
f360_statistical_tests_Check_Stationary_Hypothesis__sigma_above_threshold_test_variable_↔  
below_threshold ( )
```

Purpose:

Purpose of this test is to verify whether when sigma is above threshold and test variable is below threshold function returns true

Requirements NA.

Precondition:

Set up abs test val Set up test value std Set up test value threshold Set up sigma threshold

Action:

Call tested function

Result:

Check whether test_result is true

2.34.2.3 f360_statistical_tests_Check_Stationary_Hypothesis__sigma_and_test_variable_above_thresholds()

```
f360_statistical_tests_Check_Stationary_Hypothesis__sigma_and_test_variable_above_thresholds ( )
```

Purpose:

Purpose of this test is to verify whether when sigma is above threshold and test value is above threshold function returns false

Requirements NA.

Precondition:

Set up abs test val Set up test value std Set up test value threshold Set up sigma threshold

Action:

Call tested function

Result:

Check whether test_result is false

2.34.2.4 f360_statistical_tests_Check_Stationary_Hypothesis__sigma_below_threshold_test_variable_above_threshold()

```
f360_statistical_tests_Check_Stationary_Hypothesis__sigma_below_threshold_test_variable_↵  
above_threshold ( )
```

Purpose:

Purpose of this test is to verify whether when sigma is below threshold and test variable is above threshold function returns true

Requirements NA.

Precondition:

Set up abs test val Set up test value std Set up test value threshold Set up sigma threshold

Action:

Call tested function

Result:

Check whether test_result is true

2.34.2.5 f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_calculated()

```
f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_calculated (   
 )
```

Purpose:

Purpose of this test is to verify whether F360_Chi_Square_Test_Two_Degree_P_Value properly calculates p_value

Requirements NA.

Precondition:

Set up test variable Set up maximum test variable Set up expected probability value

Action:

Call tested function

Result:

Check whether calculated probability value is equal to expected value.

2.34.2.6 f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_saturated()

```
f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_saturated ( )
```

Purpose:\n

Purpose of this test is to verify whether F360_Chi_Square_Test_Two_Degree_P_Value properly calculates p_value when its above 1 and should be saturated.

Requirements NA.

Precondition:\n

Set up test variable such that initially calculated p_value is > 1 Set up maximum test variable > test variable
Set up expected probability value to 1

Action:\n

Call tested function

Result:\n

Check whether calculated probability value is equal to expected value.

2.34.2.7 f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__test_variable_greater_than_max_test_varia

```
f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__test_variable_greater_than_↵  
max_test_variable ( )
```

Purpose:\n

Purpose of this test is to verify whether when test variable is greater than maximum test variable, minimum probability is returned

Requirements NA.

Precondition:\n

Set up test variable Set up maximum test variable Set up expected probability value

Action:\n

Call tested function

Result:\n

Check whether calculated probability value is equal to expected value.

2.34.2.8 f360_statistical_tests_F360_Try_To_Reject_Test__properly_does_not_reject_hypothesis()

```
f360_statistical_tests_F360_Try_To_Reject_Test__properly_does_not_reject_hypothesis ( )
```

Purpose:

Purpose of this test is to verify whether test properly does not reject hypothesis under test when p_value is greater or equal given significance level alpha

Requirements NA.

Precondition:

Set up p_value Set up alpha level

Action:

Call tested function

Result:

Check whether test_result is false

2.34.2.9 f360_statistical_tests_F360_Try_To_Reject_Test__properly_rejects_hypothesis()

```
f360_statistical_tests_F360_Try_To_Reject_Test__properly_rejects_hypothesis ( )
```

Purpose:

Purpose of this test is to verify whether test properly rejects hypothesis under test when p_value is below given significance level alpha

Requirements NA.

Precondition:

Set up p_value Set up alpha level

Action:

Call tested function

Result:

Check whether test_result is true

2.35 Rspp_try_to_dealias_range_rate

Functions

- test group [rspp_try_to_dealias_range_rate](#) ()

Test group contains common variable declaration for all tests. The sensor properties are defined in this test group and is the same for all tests. To be able to dealias range rates towards a prediction a threshold is needed which is the same for all tests and choosen so that it's easy to derive expected data for different test cases.

- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Base_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Second_Lower_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Third_Upper_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_Second](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_First_Upper](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fifth_Lower](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fourth_Lower](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Both_In_Second](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Exactly_On_Limit](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_Base](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_First_Upper](#) ()

2.35.1 Detailed Description

2.35.2 Function Documentation

2.35.2.1 rspp_try_to_dealias_range_rate()

```
test group rspp_try_to_dealias_range_rate ( )
```

Test group contains common variable declaration for all tests. The sensor properties are defined in this test group and is the same for all tests. To be able to dealias range rates towards a prediction a threshold is needed which is the same for all tests and choosen so that it's easy to derive expected data for different test cases.

Setup:\n

Setup arbitrary sensor settings

2.35.2.2 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Base_Interval()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Base_Interval ( )
```

Purpose:

Verify that range rate is dealiased correctly when measured range rate and predicted range rate is in the base interval.

Requirements NA

Precondition:

Set range rate and predicted range rate to 0

Action:

Call function

Result:

Compare against expected data

2.35.2.3 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Both_Interval()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_↵  
Upper_Both_In_Second ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is in the second upper interval and close to the first upper interval. Measured range rate is in the second upper interval and close to the first interval limit.

Requirements NA

Precondition:

Set predicted range rate in the second lower interval. Set measured range rate to match measured aliased range rate if the true range rate was in the second lower interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.4 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_↵  
Upper_Pred_In_First_Upper ( )
```

Purpose:\n

Verify that range rate is dealiased correctly when predicted range rate is in the first upper interval and close to the second interval. Measured range rate is in the second interval and close to the interval limit.

Requirements NA

Precondition:\n

Set predicted range rate in the first upper interval. Set measured range rate to match measured aliased range rate if the true range rate was in the second upper interval.

Action:\n

Call function

Result:\n

Compare against expected data

2.35.2.5 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_↵  
Upper_Pred_In_Second_Upper ( )
```

Purpose:\n

Verify that range rate is dealiased correctly when predicted range rate is in the second upper interval and close to the interval limit. Measured range rate is in the first interval and close to the interval limit.

Requirements NA

Precondition:\n

Set predicted range rate in the second upper interval. Set measured range rate to match measured aliased range rate if the true range rate was in the first upper interval.

Action:\n

Call function

Result:\n

Compare against expected data

2.35.2.6 rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In

```
rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_↵  
Lower_Pred_In_Fifth_Lower ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is in the fifth lower interval and close to the fourth interval. Measured range rate is in the fourth lower interval and close to the interval limit.

Requirements NA

Precondition:

Set predicted range rate in the fifth lower interval. Set measured range rate to match measured aliased range rate if the true range rate was in the fourth lower interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.7 rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In

```
rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Between_Fourth_And_Fifth_↵  
Lower_Pred_In_Fourth_Lower ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is in the fourth lower interval and close to the fifth interval. Measured range rate is in the fifth lower interval and close to the interval limit.

Requirements NA

Precondition:

Set predicted range rate in the fifth lower interval. Set measured range rate to match measured aliased range rate if the true range rate was in the fourth lower interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.8 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is exactly on the interval limit between first and second lower interval. Measured range rate exactly on the min aliased range rate of the sensor.

Requirements NA

Precondition:

Set predicted range rate on the interval limit between first and second lower interval. Set measured range rate to match measured aliased range rate if the true range rate was in the first lower interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.9 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Exactly_On_Limit()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Exactly_On_Limit ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is exactly on the interval limit between first and second lower interval. Measured range rate is in the second lower interval and close to the first interval limit.

Requirements NA

Precondition:

Set predicted range rate on the interval limit between first and second upper interval. Set measured range rate to match measured aliased range rate if the true range rate was in the second upper interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.10 rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_Base()

```
rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_Base ( )
```

Purpose:\n

Verify that range rate is not dealiased when predicted range rate doesn't match measured dealiased range rate. Prediction in base interval and measured also in base interval but not matching prediction.

Requirements NA**Precondition:**\n

Set predicted range rate in the base interval. Set measured range rate in the base interval but different from prediction

Action:\n

Call function

Result:\n

Compare against expected data

2.35.2.11 rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_First_Upper()

```
rspp_try_to_dealiase_range_rate_Try_To_Dealiase_Range_Rate_Test_Pred_Not_Matching_First_Upper  
( )
```

Purpose:\n

Verify that range rate is not dealiased when predicted range rate doesn't match measured dealiased range rate. Prediction in first upper interval and measured in base interval and does not match prediction.

Requirements NA**Precondition:**\n

Set predicted range rate in first upper interval. Set measured range rate in the base interval but different from prediction

Action:\n

Call function

Result:\n

Compare against expected data

2.35.2.12 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Second_Lower_Interval()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Second_Lower_Interval ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is in the second lower interval and measured range rate matches prediction and has been aliased.

Requirements NA

Precondition:

Set predicted range rate in the second lower interval. Set measured range rate to match measured aliased range rate if the true range rate was in the second lower interval.

Action:

Call function

Result:

Compare against expected data

2.35.2.13 rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Third_Upper_Interval()

```
rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Third_Upper_Interval ( )
```

Purpose:

Verify that range rate is dealiased correctly when predicted range rate is in the third upper interval and measured range rate matches prediction and has been aliased.

Requirements NA

Precondition:

Set predicted range rate in the third upper interval. Set measured range rate to match measured aliased range rate if the true range rate was in the third upper interval.

Action:

Call function

Result:

Compare against expected data

2.36 F360_uncertainty_propagation

Functions

- test group [f360_uncertainty_propagation](#) ()
Add brief description of test group.
- [f360_uncertainty_propagation_Test_Constant_Uncertainty_Propagation_2d](#) ()

2.36.1 Detailed Description

2.36.2 Function Documentation

2.36.2.1 f360_uncertainty_propagation()

```
test_group f360_uncertainty_propagation ( )
```

Add brief description of test group.

Setup:\n

Set up thresholds for comparing if floats are equal

Teardown:\n

Nothing to teardown in this test group

2.36.2.2 f360_uncertainty_propagation_Test_Constant_Uncertainty_Propagation_2d()

```
f360_uncertainty_propagation_Test_Constant_Uncertainty_Propagation_2d ( )
```

Purpose:\n Purpose of test is to verify that function `Test_Constant_Uncertainty_Propagation_2d`

works as intended. Intended functionality is that function takes a constant and a 2D covariance matrix as input and multiplies each element of the matrix with the squared constant.

Requirements NA

Step 1:\n

Check that functions returns a zero matrix when it gets a zero matrix as input

Precondition:\n

Set up a non-zero constant and a zero input matrix and an expected output matrix

Action:\n

Call the function to generate the output matrix

Result:\n

Check so that all elements of matrix is equal to the expected result

Step 2:\n

Check that functions returns a zero matrix when it gets a zero constant as input

Precondition:\n

Set up a zero constant and a non-zero input matrix and an expected output matrix

Action:\n

Call the function to generate the output matrix

Result:\n

Check so that all elements of matrix is equal to the expected result

Step 3:\n

Check that functions returns correct matrix when it gets a positive constant and a non-zero matrix

Precondition:\n

Set up a zero constant and a non-zero input matrix and an expected output matrix

Action:\n

Call the function to generate the output matrix

Result:\n

Check so that all elements of matrix is equal to the expected result

Step 4:\n

Check that functions returns correct matrix when it gets a negative constant and a non-zero matrix

Precondition:\n

Set up a zero constant and a non-zero input matrix and an expected output matrix

Action:\n

Call the function to generate the output matrix

Result:\n

Check so that all elements of matrix is equal to the expected result

2.37 F360_update_sensor_first_detection_index

Functions

- test group [f360_update_sensor_first_detection_index](#) ()
Test that the functionality of Update_Sensor_First_Detection_Index works as intended.
- [f360_update_sensor_first_detection_index_Update_Sensor_First_Detection_Index](#) ()
- [f360_update_sensor_first_detection_index_update_sensor_first_detection_index_too_many_detections](#) ()

2.37.1 Detailed Description

2.37.2 Function Documentation

2.37.2.1 f360_update_sensor_first_detection_index()

```
test_group f360_update_sensor_first_detection_index ( )
```

Test that the functionality of Update_Sensor_First_Detection_Index works as intended.

Setup:\n

Set up a list of raw detections coming from different sensors.

2.37.2.2 f360_update_sensor_first_detection_index_Update_Sensor_First_Detection_Index()

```
f360_update_sensor_first_detection_index_Update_Sensor_First_Detection_Index ( )
```

Purpose:\n Tests that Update_Sensor_First_Detection_Index correctly updates first_detection_list_idx for each sensor.

\

Requirements NA

Precondition:\n

Set up expected data for sensor first detection indices.

Action:\n

Call Update_Sensor_First_Detection_Index().

Result:\n

Check that the first detection indices in sensor props corresponds to the expected data.

2.37.2.3 f360_update_sensor_first_detection_index_update_sensor_first_detection_index_too_many_detections()

```
f360_update_sensor_first_detection_index_update_sensor_first_detection_index_too_many_detections  
( )
```

Purpose: Tests that Update_Sensor_First_Detection_Index correctly updates first_detection_list_idx even when number_of_valid_detections is large
\\

Requirements NA

Precondition:

Set up expected data for sensor first detection indices. raw_detection_list.number_of_valid_detections is set higher than it actually is. (Likely not possible during normal code execution, but needed to reach full test coverage.)

Action:

Call Update_Sensor_First_Detection_Index().

Result:

Check that the first detection indices in sensor props corresponds to the expected data.

2.38 F360_update_sensors_fov

Functions

- test group [f360_update_sensors_fov](#) ()
Tests to check the updated sensor properties with moving host and the azimuth properties within and beyond fov.
- [f360_update_sensors_fov_update_sensors_fov](#) ()
- [f360_update_sensors_fov_update_sensors_fov_azimuth_beyond_fov](#) ()

2.38.1 Detailed Description

2.38.2 Function Documentation

2.38.2.1 f360_update_sensors_fov()

```
test group f360_update_sensors_fov ( )
```

Tests to check the updated sensor properties with moving host and the azimuth properties within and beyond fov.

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.38.2.2 f360_update_sensors_fov_update_sensors_fov()

```
f360_update_sensors_fov_update_sensors_fov ( )
```

Purpose:\n Check if the updated sensor properties, when the host is moving are as desired

Requirements NA

Precondition:\n

Initialize the host properties, and all the radar properties along with the calibrations

Action:\n

update the sensor_properties such as fov properties, velocity and position

Result:\n

check if the updated properties due to moving host vehicle are as desired

2.38.2.3 f360_update_sensors_fov_update_sensors_fov_azimuth_beyond_fov()

```
f360_update_sensors_fov_update_sensors_fov_azimuth_beyond_fov ( )
```

Purpose:\n Check for updated sensor properties, when host is moving and azimuth is set beyond fov.

Requirements NA

Precondition:\n

Initialize the host properties, and all the radar properties along with the calibrations

Action:\n

update the sensor_properties such as its fov properties its velocity and position

Result:\n

check if the updated properties are at the fov limit and not beyond it.

2.39 F360_update_sensors_time

Functions

- test group `f360_update_sensors_time` ()
Test to see if the time_since_measurement is updated.
- `f360_update_sensors_time_update_time_since_measurement_in_sensor_props` ()

2.39.1 Detailed Description

2.39.2 Function Documentation

2.39.2.1 f360_update_sensors_time()

```
test group f360_update_sensors_time ( )
```

Test to see if the time_since_measurement is updated.

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.39.2.2 f360_update_sensors_time_update_time_since_measurement_in_sensor_props()

```
f360_update_sensors_time_update_time_since_measurement_in_sensor_props ( )
```

Purpose:\n Check if time_since_measurement for each sensor is updated

Precondition:\n

setup tracker time and sensor timestamps

Action:\n

update time since measurement in the sensor properties

Result:\n

Check the time since measurement for different sensors

2.40 F360_update_sensors

Functions

- test group `f360_update_sensors` ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- `f360_update_sensors_FunctionToTest_Descriptive_Tag` ()

2.40.1 Detailed Description

2.40.2 Function Documentation

2.40.2.1 `f360_update_sensors()`

```
test group f360_update_sensors ( )
```

Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.

Setup:\n

Describe what is done in test setup. Remove test setup function and this tag if it is not used.

Teardown:\n

Describe what is done in test teardown. Remove test teardown function and this tag if it is not used.

2.40.2.2 `f360_update_sensors_FunctionToTest_Descriptive_Tag()`

```
f360_update_sensors_FunctionToTest_Descriptive_Tag ( )
```

Purpose:\n

Describe purpose of test on a high level. The description should be unique and distinguishable from other tests' purposes. E.g. test that function `Some_Function()` works as intended given some specific preconditions.

Requirements Put in requirement tag if any otherwise set to NA.

Precondition:\n

Describe prepared input. E.g. describe relevant set up from test group and what variables are changed or added for this specific test.

Action:\n

Describe the action of the test. E.g. call `Some_Function()`.

Result:\n

Describe expected output. E.g. check that the output match expected data.

2.41 F360_update_sensors_valid_info

Functions

- test group `f360_update_sensors_valid_info()`
A group of tests that check if valid sensors are updated in the tracker_info struct.
- `f360_update_sensors_valid_info_check_esr_valid()`
- `f360_update_sensors_valid_info_check_srr_valid()`
- `f360_update_sensors_valid_info_check_srr_esr_valid()`
- `f360_update_sensors_valid_info_check_invalid_sensor()`

2.41.1 Detailed Description

2.41.2 Function Documentation

2.41.2.1 f360_update_sensors_valid_info()

```
test_group f360_update_sensors_valid_info ( )
```

A group of tests that check if valid sensors are updated in the tracker_info struct.

Requirements : NA

Setup:\n

Nothing to setup in this test group

Teardown:\n

Nothing to teardown in this test group

2.41.2.2 f360_update_sensors_valid_info_check_esr_valid()

```
f360_update_sensors_valid_info_check_esr_valid ( )
```

Purpose:\n Checking if valid ESR sensors are updated in the tracker_info struct

Requirements : NA

Precondition:\n

Set valid ESR sensor

Action:\n

Update the valid sensor info in the tracker_info

Result:\n

Check if valid ESR Sensor is updated

2.41.2.3 f360_update_sensors_valid_info_check_invalid_sensor()

```
f360_update_sensors_valid_info_check_invalid_sensor ( )
```

Purpose: Checking at boundary conditions when the sensors are not valid

Requirements : NA

Precondition:

Set invalid SRR and ESR sensors

Action:

Update the invalid sensors in the tracker_info

Result:

Check SRR and ESR Sensors are updated as invalid

2.41.2.4 f360_update_sensors_valid_info_check_srr_esr_valid()

```
f360_update_sensors_valid_info_check_srr_esr_valid ( )
```

Purpose: Checking if valid SRR and ESR sensors are updated in the tracker_info struct

Requirements : NA

Precondition:

Set valid SRR and ESR sensors

Action:

Update the valid sensor info in the tracker_info

Result:

Check if Valid SRR and ESR Sensors are updated

2.41.2.5 f360_update_sensors_valid_info_check_srr_valid()

```
f360_update_sensors_valid_info_check_srr_valid ( )
```

Purpose:\n Checking if valid SRR sensors are updated in the tracker_info struct

Requirements : NA

Precondition:\n

Set valid SRR sensor

Action:\n

Update the valid sensor info in the tracker_info

Result:\n

Check if valid SRR Sensor is updated

2.42 F360_clear_dets_vcs_long_sorted_info

Functions

- test group `f360_clear_dets_vcs_long_sorted_info()`
Tests to ensure that function Clear_Dets_Vcs_Long_Sorted_Info works as expected.
- `f360_clear_dets_vcs_long_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero()`

2.42.1 Detailed Description

2.42.2 Function Documentation

2.42.2.1 f360_clear_dets_vcs_long_sorted_info()

```
test_group f360_clear_dets_vcs_long_sorted_info ( )
```

Tests to ensure that function Clear_Dets_Vcs_Long_Sorted_Info works as expected.

Setup:\n

Declare raw_detect_list structure

2.42.2.2 f360_clear_dets_vcs_long_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero()

```
f360_clear_dets_vcs_long_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_↵  
Over_Zero ( )
```

Purpose:\n Purpose is to test that data in raw detection list related to longitudinal reference points are cleared
correctly.

Requirements NA

Action:\n

Call function

Result:\n

Check that all relevant fields have been cleared

2.43 F360_update_dets_vcs_long_ref_sorted_info

Functions

- test group [f360_update_dets_vcs_long_ref_sorted_info](#) ()
Tests for Update_Dets_Vcs_Long_Ref_Sorted_Info(). Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() in all possible branches and extreme situations when detection is below/above all reference points in vcslong_sorted_ref_det_idx to an arbitrary valid index as this is expected to be untouched.
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#) ()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero_Prev_Re](#) ()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_All_Ref_Points](#) ()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_All_Ref_Points](#) ()

2.43.1 Detailed Description

2.43.2 Function Documentation

2.43.2.1 f360_update_dets_vcs_long_ref_sorted_info()

```
test_group f360_update_dets_vcs_long_ref_sorted_info ( )
```

Tests for Update_Dets_Vcs_Long_Ref_Sorted_Info(). Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() in all possible branches and extreme situations when detection is below/above all reference points in vcslong_sorted_ref_det_idx to an arbitrary valid index as this is expected to be untouched.

Setup:\n

Declare and setup common data for all test cases Set detection vcs long position greater than zero and give it arbitrary index Also set first and last element in raw_detect_list.vcslong_sorted_ref_det_idx which is expected to be untouched

2.43.2.2 f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_A

```
f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_All_Ref_Points ( )
```

Purpose:\n Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() when detection is lower than all reference points.

We thus expect all reference indexes to be invalid

Requirements NA

Precondition:\n

Set detection long position to lower than lowest reference point defined in calibration

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.43.2.3 f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_All

```
f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_↔
Det_Over_All_Ref_Points ( )
```

Purpose:\n Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() when detection is above all reference points.

We thus expect all reference indexes to have the index of the detection

Requirements NA

Precondition:\n

Set detection long position to greater than highest reference point defined in calibration

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.43.2.4 f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero

```
f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_↔
Det_Over_Zero ( )
```

Purpose:\n Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() when all reference indexes are invalid

Requirements NA

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.43.2.5 f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zer

```
f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zer_Prev_Ref_Point_Exist ( )
```

Purpose:\n Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() when some reference indexes already have been filled

Requirements NA

Precondition:\n

Set the second and third reference indexes in raw detection list to an arbitrary index

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.44 F360_sort_detections_vcs_long

Functions

- test group [f360_sort_detections_vcs_long](#) ()
Tests for Sort_Detections_Vcs_Long.
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_General](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_One_Detection](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_No_Detection](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point](#) ()

2.44.1 Detailed Description

2.44.2 Function Documentation

2.44.2.1 f360_sort_detections_vcs_long()

```
test group f360_sort_detections_vcs_long ( )
```

Tests for Sort_Detections_Vcs_Long.

Setup:\n

Declare and setup common data for several tests

2.44.2.2 f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point()

```
f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point  
( )
```

Purpose:\n Purpose is to test function Sort_Detections_Vcs_Long() in the case when a detection is larger than the largest reference point

Requirements NA

Precondition:\n

Set longitudinal position of largest vcs-long detection larger than the largest reference point

Action:\n

Call function

Result:\n

Check expected data in raw detection list and detection props structure

2.44.2.3 f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_General()

```
f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_General ( )
```

Purpose:\n Purpose is to test function Sort_Detections_Vcs_Long() in the general case where there are more than 1 detections

Requirements NA

Action:\n

Call function

Result:\n

Check expected data in raw detection list and detection props

2.44.2.4 f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_No_Detection()

```
f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_No_Detection ( )
```

Purpose:\n Purpose is to test function Sort_Detections_Vcs_Long() in the case when there are no detections

Requirements NA

Precondition:\n

Set number of detections to 0

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.44.2.5 f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_One_Detection()

```
f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_One_Detection ( )
```

Purpose:\n Purpose is to test function Sort_Detections_Vcs_Long() in the case when there are only 1 detection

Requirements NA

Precondition:\n

Set number of detections to 1

Action:\n

Call function

Result:\n

Check expected data in raw detection list and detection props

2.45 F360_get_first_relevant_long_sorted_det_idx

Functions

- test group `f360_get_first_relevant_long_sorted_det_idx()`
Tests for Get_First_Relevant_Long_Sorted_Det_Idx.
- `f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_General_Case()`
- `f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_Desired_Value()`
- `f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_Desired_Value2()`

2.45.1 Detailed Description

2.45.2 Function Documentation

2.45.2.1 f360_get_first_relevant_long_sorted_det_idx()

```
test_group f360_get_first_relevant_long_sorted_det_idx ( )
```

Tests for Get_First_Relevant_Long_Sorted_Det_Idx.

Setup:\n

Declare and setup common data for several tests

2.45.2.2 f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_General_Case()

```
f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_↵  
General_Case ( )
```

Purpose:\n Purpose is to test function Get_First_Relevant_Long_Sorted_Det_Idx() in the general case when

a detection exists above one of the reference points in raw detection list

Requirements NA

Precondition:\n

Set desired vcs_long_value

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.45.2.3 f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_D

```
f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No↔  
Dets_Above_Desired_Value ( )
```

Purpose:\n Purpose is to test function Get_First_Relevant_Long_Sorted_Det_Idx() in the case when
there are no detections above desired value

Requirements NA

Precondition:\n

Set desired vcs_long_value

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

2.45.2.4 f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_D

```
f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No↔  
Dets_Above_Desired_Value_Full_Loop ( )
```

Purpose:\n Purpose is to test function Get_First_Relevant_Long_Sorted_Det_Idx() in the case when
the desired value is above the last reference point

Requirements NA

Precondition:\n

Set desired vcs_long_value and detection position to something very large to ensure the detection is last in the reference index array in raw detection list. Since we modify detection in this test we also need to sort the detections

Action:\n

Call function

Result:\n

Check expected data in raw detection list.

Chapter 3

File Documentation

3.1 source/unittest/rspp_calibrations_unittest.cpp File Reference

Tests

- test group [rspp_calibrations](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [rspp_calibrations_FunctionToTest_Descriptive_Tag](#) ()

3.1.1 Detailed Description

This file contains unit tests for content of rspp_calibrations.cpp file

3.2 source/unittest/rspp_check_det_az_conf_and_elevation_qualtest_↵ unittest.cpp File Reference

Tests

- test group [f360_check_det_az_conf_and_elevation_qualtest](#) ()
This test group marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle.
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation1](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation2](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_Check_Det_Az_Conf_and_Elevation3](#) ()
- [f360_check_det_az_conf_and_elevation_qualtest_f360_check_det_az_conf_and_elevation_off](#) ()

3.2.1 Detailed Description

This test file marks out unassociated detections that are suspicious based on their azimuth confidence value and elevation angle

3.3 source/unittest/rspp_check_det_az_conf_and_elevation_unittest.cpp File Reference

Tests

- test group [f360_check_det_az_conf_and_elevation](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [f360_check_det_az_conf_and_elevation_FunctionToTest_Descriptive_Tag](#) ()

3.3.1 Detailed Description

This file contains unit tests for content of f360_check_det_az_conf_and_elevation.cpp file

3.4 source/unittest/rspp_compute_detections_properties_unittest.cpp File Reference

Tests

- test group [compute_detection_properties](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [compute_detection_properties_Compute_Detection_VCS_Coordinates](#) ()
- [compute_detection_properties_Test_Updated_detection_Time_with_data](#) ()
- [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_positive_polarity](#) ()
- [compute_detection_properties_Compensate_Detection_VCS_Azimuth_And_Elevation_Misalignment_Angle_negative_polarity](#) ()

3.4.1 Detailed Description

This file contains unit tests for content of rspp_compute_detections_properties.cpp file

3.5 source/unittest/rspp_detection_double_bounce_detector_qualtest_unittest.cpp File Reference

Tests

- test group [rspp_detection_double_bounce_detector_qualtest](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_true](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RangeMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_AzimuthMismatch](#) ()
- [rspp_detection_double_bounce_detector_qualtest_DoubleBounce_false_RdotMismatch](#) ()
- test group [system_test_rspp_detection_double_bounce_detector](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [system_test_rspp_detection_double_bounce_detector_Double_Bounce_True](#) ()
- [system_test_rspp_detection_double_bounce_detector_Multiple_Double_Bounce_True](#) ()
- [system_test_rspp_detection_double_bounce_detector_Arbitrary_Double_Bounce_Scenario](#) ()

3.5.1 Detailed Description

This file contains basic unit tests for algorithm that flags detections as double bounce if they fit the criteria.

3.6 source/unittest/rspp_detection_double_bounce_detector_unittest.cpp File Reference

Tests

- test group [rspp_detection_double_bounce_detector](#) ()
Basic test to verify that detection gets flagged as double bounce when expected and vice versa.
- [rspp_detection_double_bounce_detector_Test_with_2_detection_with_double_bounce](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_4](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_3](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_2](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_outside_gate_1](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_inside_gate](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_3](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_2](#) ()
- [rspp_detection_double_bounce_detector_Test_with_2_detection_not_in_limit_1](#) ()
- [rspp_detection_double_bounce_detector_Test_one_detection_per_sensor](#) ()

3.6.1 Detailed Description

This file contains basic unit tests for [rspp_detection_double_bounce_detector_unittest.cpp](#)

3.7 source/unittest/rspp_detection_multipath_filter_unittest.cpp File Reference

Tests

- test group [f360_detection_multipath_filter](#) ()
Test that the functionality in Filter_Out_Multipath_Detections() works as intended.
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Filter_First_Det_Idx_Invalid](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Filter_Num_Bad_Dets_Above_Thresh](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Filter_Num_Bad_Dets_Below_Thresh](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Num_Valid_Dets_Below_Threshold](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Host_Slow](#) ()
- [f360_detection_multipath_filter_Detection_Multipath_Filter_Not_SRR5](#) ()

3.7.1 Detailed Description

This file contains unit tests for content of [f360_detection_multipath_filter.cpp](#) file

3.8 [source/unittest/rspp_discard_detections_outside_sensor_fov_↔](#) unittest.cpp File Reference

Tests

- test group [rspp_discard_detections_outside_sensor_fov](#) ()
This test group verifies the functionality of Discard_Detections_Outside_Sensor_Field_Of_View.
- [rspp_discard_detections_outside_sensor_fov_All_Detections_Inside_FoV](#) ()
- [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_Based_On_Their_Sensor_Id](#) ()
- [rspp_discard_detections_outside_sensor_fov_All_Detections_But_One_Outside_FoV](#) ()
- [rspp_discard_detections_outside_sensor_fov_Check_Detections_Are_Flagged_As_Per_Sensor_Lookid](#) ()

3.8.1 Detailed Description

This file contains unit tests for content of rspp_discard_detections_outside_sensor_fov.cpp file

3.9 [source/unittest/rspp_ego_motion_compensate_dets_qualtest_↔](#) unittest.cpp File Reference

Tests

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host](#) ()
Check correctness of detection motion status determination when host does not move.
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_det](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__stopped_host_Detection_Motion_Classification__mark_low](#) ()
- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma](#) ()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__moving_forward_host_Detection_Motion_Classification__ma](#) ()
- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host](#) ()

Check correctness of detection motion status determination when host moves forward with high speed (speed = 30 m/s). Predicted range rate is then high enough to try to dealiase detection.

- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__fast_moving_forward_host_Detection_Motion_Classification](#)
()

3.9.1 Detailed Description

This file contains unit tests for content of `f360_Compute_Detections_Properties_Based_On_Ego_Motion.cpp` file

3.10 source/unittest/rspp_ego_motion_compensate_dets_unittest.cpp File Reference

Tests

- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__f_esr_mr_in_lr_fov](#) ()
Check if `f_esr_mr_in_lr_fov` is calculatted correctly.
- test group [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation](#) ()
Check if the threshold limitation introduced to handle low speed crossing vehicles is working as intended. Detections need to be able to get classified as moving even with low compensated range-rate in order to have acceptable tracking. However, when the host is moving a high speed or during high curvature the threshold should remain high.
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_Compute_Detections_Properties_Base](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_within_inte](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_below_inte](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_above_inte](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_if_infinite](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_at_upper](#)
()
- [f360_Compute_Detections_Properties_Based_On_Ego_Motion__threshold_limitation_calculate_range_rate_comp_at_lower](#)
()
- test group [f360_ego_motion_compensate_dets__stopped_host_qualtest](#) ()
Check correctness of detection motion status determination when host does not move.
- [IGNORE_TEST \(f360_ego_motion_compensate_dets__stopped_host_qualtest, Ego_Motion_Compensate↔](#)
[_Dets__det_rr_comp\)](#)
- test group [f360_ego_motion_compensate_dets__moving_forward_host_qualtest](#) ()
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [IGNORE_TEST \(f360_ego_motion_compensate_dets__moving_forward_host_qualtest, Detection_Motion↔](#)
[_Classification__det_rr_comp_calculation\)](#)

3.10.1 Detailed Description

This file contains unit tests for content of f360_Compute_Detections_Properties_Based_On_Ego_Motion.cpp file

3.11 source/unittest/rspp_filter_detections_with_bad_azimuth_confidence_unittest.cpp File Reference

Tests

- test group [f360_filter_detections_with_bad_azimuth_confidence](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [f360_filter_detections_with_bad_azimuth_confidence_FunctionToTest_Descriptive_Tag](#) ()

3.11.1 Detailed Description

This file contains unit tests for content of f360_filter_detections_with_bad_azimuth_confidence.cpp file

3.12 source/unittest/rspp_get_version_unittest.cpp File Reference

Tests

- test group [rspp_get_version](#) ()
Check that RSPP_Get_Version returns the expected values.
- [rspp_get_version_RSPP_Get_Version_Output_Test](#) ()

3.12.1 Detailed Description

This file contains unit tests for content of rspp_get_version.cpp file

3.13 source/unittest/rspp_host_props_update_unittest.cpp File Reference

Tests

- test group [f360_host_props_update](#) ()
Test group for Host_Props_Update. Tests in this test group verify that the host properties are time updated correctly.
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Heading_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Covariance_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_X_Position_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Heading_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Covariance_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Driving_Straight_Alone_Y_Position_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Covariance_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Non_Zero_Sideslip_Position_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Yawing_Heading_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Yawing_Covariance_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Yawing_Position_Update](#) ()
- [f360_host_props_update_Host_Props_Update_Yawing_Heading_Normalized](#) ()

3.13.1 Detailed Description

This file contains unit tests for content of f360_host_props_update.cpp file

3.14 source/unittest/rspp_input_sanity_check_unittest.cpp File Reference

Tests

- test group [rspp_input_sanity_check \(\)](#)
Testing behaviour of the RSPP Input_Sanity_Check() function.
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Valid_Detection \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_No_Detection \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Two_Detections \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Detection \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Invalid_Sensor_Id \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_Low_Interval_Width \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Too_High_Interval_Width \(\)](#)
- [rspp_input_sanity_check_Input_Sanity_Check_Test_With_Sensor_Valid \(\)](#)

3.14.1 Detailed Description

This file contains unit tests for content of rspp_input_sanity_check.cpp file

3.15 source/unittest/rspp_inputs_preprocessing_qualtest_unittest.cpp File Reference

Tests

- test group [inputs_preprocessing_qualtest \(\)](#)
Add brief description of test group.
- [inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements \(\)](#)
- test group [f360_ego_motion_compensate_dets__stopped_host_qualtest \(\)](#)
Check correctness of detection motion status determination when host does not move.
- [f360_ego_motion_compensate_dets__stopped_host_qualtest_Ego_Motion_Compensate_Dets__det_rr_comp \(\)](#)
- test group [f360_ego_motion_compensate_dets__moving_forward_host_qualtest \(\)](#)
Check correctness of detection motion status determination when host moves forward (speed = 10 m/s)
- [f360_ego_motion_compensate_dets__moving_forward_host_qualtest_Detection_Motion_Classification__det_rr_comp_calculat \(\)](#)

3.15.1 Detailed Description

This file contains sw qualifications tests for f360_internal_preprocessing

3.15.2 Tests

3.15.2.1 `inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements()`

```
inputs_preprocessing_qualtest_Internal_Preprocessing_Input_Interface_Requirements ( )
```

Purpose:\n Test input interface requirements.

Requirements FTCP-8256, FTCP-7799

Precondition:\n

Initialize structures of the correct type as specified in requirements.

Action:\n

Call function to test that inputs, as specified in requirements, are accepted.

Result:\n

Test should compile and run. No functionality is supposed to be tested in this test.

3.16 `source/unittest/rspp_inputs_preprocessing_unittest.cpp` File Reference

Tests

- test group `f360_inputs_preprocessing` ()
- `f360_inputs_preprocessing_Sensor_Capability_Host__SCM_Yaw_Rate_Std` ()

3.16.1 Detailed Description

This file contains unit tests for content of `f360_inputs_preprocessing.cpp` file

3.17 source/unittest/rspp_math_func_unittest.cpp File Reference

Tests

- test group [f360_math_func \(\)](#)
 - Tests for f360_math_func.*
 - [f360_math_func_test_f360_get_hypotenuse_negative_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_positive_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_zero_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_negative_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_positive_input \(\)](#)
 - [f360_math_func_test_f360_get_hypotenuse_squared_zero_input \(\)](#)
 - [f360_math_func_F360_Matmul_MxN_NxP \(\)](#)
 - [f360_math_func_F360_Matmul_MxN_NxP_Transpose \(\)](#)
 - [f360_math_func_F360_Saturate \(\)](#)
 - [f360_math_func_F360_Sort_fail \(\)](#)
 - [f360_math_func_F360_Sort_success \(\)](#)
 - [f360_math_func_F360_Sort_asc \(\)](#)
 - [f360_math_func_F360_Sort_sorted_values_not_modified \(\)](#)
 - [f360_math_func_F360_Sort_partial \(\)](#)
 - [f360_math_func_F360_Sort_no_sort \(\)](#)
 - [f360_math_func_F360_Sort_stability \(\)](#)
 - [f360_math_func_F360_Sort_dsc \(\)](#)
 - [f360_math_func_F360_Linear_Equation \(\)](#)
 - [f360_math_func_F360_Linear_Equation_small_den \(\)](#)
 - [f360_math_func_Test_F360_2d_Matrix_Determinant \(\)](#)
 - [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Sorted_Breakpoints \(\)](#)
 - [f360_math_func_Test_RSPP_Piecewise_Linear_Equation_Close_Breakpoints \(\)](#)
 - [f360_math_func_Test_F360_Linear_Equation_With_Saturation \(\)](#)
 - [f360_math_func_F360_Linear_Equation_With_Saturation_small_den \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_First_Order \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_neg_filt_pos_meas \(\)](#)
 - [f360_math_func_Test_F360_Low_Pass_Filter_Angle_First_Order_pos_filt_neg_meas \(\)](#)
 - [f360_math_func_Test_F360_Safe_Logf \(\)](#)
 - [f360_math_func_Test_F360_Power_Scale_Prob \(\)](#)
 - [f360_math_func_Test_F360_Accumulate_Probabilities \(\)](#)
 - [f360_math_func_Test_F360_matmul_2x2_2x1 \(\)](#)
 - [f360_math_func_Test_F360_Mod_divisor_with_decimals \(\)](#)
 - [f360_math_func_Test_F360_Mod_divisor_without_decimals \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Angle \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Pi_Over_3 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_6_Pi_Over_5 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix_With_Precalc_Coeff_For_Minus_Pi_Over_3 \(\)](#)
 - [f360_math_func_Test_Rotate_2D_Covariance_Matrix \(\)](#)
 - [f360_math_func_Test_F360_Multiply_2D_Matrix_With_Constant \(\)](#)
 - [f360_math_func_Test_F360_Translate_2D_Position \(\)](#)
- test group [f360_math_func_f360_rotate_2d_vector \(\)](#)
 - Tests for f360_math_func_f360_rotate_2d_vector.*
 - [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Zero \(\)](#)
 - [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_180 \(\)](#)
 - [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_90 \(\)](#)
 - [f360_math_func_f360_rotate_2d_vector_Test_F360_Rotate_2D_Vector_Angle_Is_Minus_90 \(\)](#)
 - [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_1 \(\)](#)

- [f360_math_func_F360_Bisquare_Weight_Should_Return_Correct_Result_For_Valid_Input_2 \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Tuning_Constant_Slightly_Smaller_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Slightly_Larger_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Greater_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Equal_To_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Valid_Result_For_Residual_Value_Slightly_Smaller_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Bisquare_Weight_Should_Return_Zero_For_Residual_Value_Slightly_Greater_Than_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Correct_Result_For_Valid_Input \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Negative_Tuning_Constant \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Equal_To_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Smaller_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Tuning_Constant_Greater_Than_Min_Denominator \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Equal_To_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Positive_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Tuning_Constant_Greater_Than_Absolute_Value_Of_Negative_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Valid_Result_For_Tuning_Constant_Smaller_Than_Absolute_Value_Of_Negative_Residual \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_Zero_For_Both_Input_Values_Equal_To_Zero \(\)](#)
- [f360_math_func_F360_Huber_Weight_Should_Return_One_For_Residual_Equal_To_Zero_And_Non_Zero_Tuning_Constant \(\)](#)
- [f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_No_Slope \(\)](#)
- [f360_math_func_Compute_Dist_From_Point_To_Line_Squared_Line_Non_Zero_Slope \(\)](#)
- [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_No_Slope \(\)](#)
- [f360_math_func_Compute_Projection_Of_Point_On_Line_Line_With_Non_Zero_Slope \(\)](#)
- [f360_math_func_Test_F360_Cross_Product_Of_Vectors \(\)](#)
- [f360_math_func_Test_F360_Cross_Product_Of_Negative_Vectors \(\)](#)
- [f360_math_func_Test_F360_Cross_Product_Of_Parallel_Vectors \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Positive_K_Zero_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Negative_K_Positive_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Negative_Infinite_K_Zero_M \(\)](#)
- [f360_math_func_Get_Vector_As_Linear_Equation_Positive_Infinite_K_Positive_M \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Two_Lines_Parallel_Lines \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Two_Lines_Intersection_Exists \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Imaginary_Root \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Negative_A_Coefficient \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_Small_Positive_A_Coefficient \(\)](#)
- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Small_Slope \(\)](#)

- [f360_math_func_Find_X_Intersect_Between_Line_And_2_Deg_Poly_A_Negative_One_B_Zero_Line_Large_Slope\(\)](#)
- test group [clamp\(\)](#)

Test group of Clamp function. Tests verify whether function properly clamps variable into provided limits.

3.17.1 Detailed Description

Tests for `f360_math_func`

3.18 source/unittest/rspp_norm_heading_angle_unittest.cpp File Reference

Tests

- test group [f360_norm_heading_angle\(\)](#)
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [f360_norm_heading_angle_FunctionToTest_Descriptive_Tag\(\)](#)

3.18.1 Detailed Description

This file contains unit tests for content of `f360_norm_heading_angle.cpp` file

3.19 source/unittest/rspp_sensor_capability_unittest.cpp File Reference

Tests

- test group [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate\(\)](#)
This test group tests the functionality of the function `Get_Uncertainty_Of_Compensated_Range_Rate()`
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Inside_Range\(\)](#)
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_On_Boundary_Of_Range\(\)](#)
- [f360_sensor_capability_Get_Uncertainty_Of_Compensated_Range_Rate_Test_Cos_Sin_Az_Out_Of_Range\(\)](#)
- test group [f360_sensor_capability_Compute_Raw_Detection_Uncertainty\(\)](#)
This test group tests the functionality of the function `Compute_Raw_Detection_Uncertainty()`
- [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty\(\)](#)
- [f360_sensor_capability_Compute_Raw_Detection_Uncertainty_Test_Compute_Raw_Detection_Uncertainty_2\(\)](#)
- test group [f360_sensor_capability_Get_Host_Velocity_Uncertainty\(\)](#)
This test group tests the functionality of the function `Get_Host_Velocity_Uncertainty()`
- [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_Forward_speed\(\)](#)
- [f360_sensor_capability_Get_Host_Velocity_Uncertainty_Test_with_backward_speed\(\)](#)

- test group [f360_sensor_capability_Sensor_Capability_Detections \(\)](#)
This test group tests the functionality of the function `Sensor_Capability_Detections()`
- [f360_sensor_capability_Sensor_Capability_Detections_Test_Sensor_Capability_Detections \(\)](#)
- test group [f360_sensor_capability_Sensor_Capability_Host \(\)](#)
This test group tests the functionality of the function `Sensor_Capability_Host()`
- [f360_sensor_capability_Sensor_Capability_Host_Test_Sensor_Capability_Host \(\)](#)
- test group [f360_sensor_capability \(\)](#)
This test group tests sensor_capability file.
- [f360_sensor_capability_Test_Compute_Raw_Host_Speed_Uncertainty \(\)](#)
- [f360_sensor_capability_Get_Uncertainty_Of_Detection_Position \(\)](#)
- [f360_sensor_capability_Test_Compute_Detection_Cross_Covariances \(\)](#)

3.19.1 Detailed Description

This file contains unit tests for content of `f360_sensor_capability.cpp` file

3.20 [source/unittest/rspp_set_default_flags_raw_detections_↵](#) unittest.cpp File Reference

Tests

- test group [f360_set_default_flags_raw_detections \(\)](#)
- [f360_set_default_flags_raw_detections_set_default_values_for_raw_detections \(\)](#)

3.20.1 Detailed Description

This file contains unit tests for content of `f360_set_default_flags_raw_detections.cpp` file

3.21 [source/unittest/rspp_statistical_tests_unittest.cpp](#) File Reference

Tests

- test group [f360_statistical_tests \(\)](#)
Test group of `f360_statistical_tests.cpp` file content. Tests verify whether statistical tests are properly performed.
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_calculated \(\)](#)
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__p_value_properly_saturated \(\)](#)
- [f360_statistical_tests_F360_Chi_Square_Test_Two_Degree_P_Value__test_variable_greater_than_max_test_variable \(\)](#)
- [f360_statistical_tests_F360_Try_To_Reject_Test__properly_rejects_hypothesis \(\)](#)
- [f360_statistical_tests_F360_Try_To_Reject_Test__properly_does_not_reject_hypothesis \(\)](#)
- [f360_statistical_tests_Check_Stationary_Hypothesis__handles_std_equal_to_zero \(\)](#)
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_and_test_variable_above_thresholds \(\)](#)
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_above_threshold_test_variable_below_threshold \(\)](#)
- [f360_statistical_tests_Check_Stationary_Hypothesis__sigma_below_threshold_test_variable_above_threshold \(\)](#)

3.21.1 Detailed Description

This file contains unit tests for content of f360_statistical_tests.cpp file

3.22 source/unittest/rspp_try_to_dealias_range_rate_unittest.cpp File Reference

Tests

- test group [rspp_try_to_dealias_range_rate](#) ()
 - Test group contains common variable declaration for all tests. The sensor properties are defined in this test group and is the same for all tests. To be able to dealias range rates towards a prediction a threshold is needed which is the same for all tests and choosen so that it's easy to derive expected data for different test cases.*
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Base_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Second_Lower_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Third_Upper_Interval](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_Second](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Pred_In_First_Upper](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fifth_Lower](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_Fourth_And_Fifth_Lower_Pred_In_Fourth_Lower](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Between_First_And_Second_Upper_Both_In_Second_Upper](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Exactly_On_Limit](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Both_Pred_And_Meas_Exactly_On_Limit](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_Base](#) ()
- [rspp_try_to_dealias_range_rate_Try_To_Dealias_Range_Rate_Test_Pred_Not_Matching_First_Upper](#) ()

3.22.1 Detailed Description

This file contains unit tests for content of try_to_dealias_range_rate.cpp file

3.23 source/unittest/rspp_uncertainty_propagation_unittest.cpp File Reference

Tests

- test group [f360_uncertainty_propagation](#) ()
 - Add brief description of test group.*
- [f360_uncertainty_propagation_Test_Constant_Uncertainty_Propagation_2d](#) ()

3.23.1 Detailed Description

This file contains unit tests for all functions contained in f360_uncertainty_propagation

3.24 [source/unittest/rspp_update_sensor_first_detection_index_↔](#) unittest.cpp File Reference

Tests

- test group [f360_update_sensor_first_detection_index](#) ()
Test that the functionality of Update_Sensor_First_Detection_Index works as intended.
- [f360_update_sensor_first_detection_index_Update_Sensor_First_Detection_Index](#) ()
- [f360_update_sensor_first_detection_index_update_sensor_first_detection_index_too_many_detections](#) ()

3.24.1 Detailed Description

This file contains unit tests for content of f360_update_sensor_first_detection_index.cpp file

3.25 [source/unittest/rspp_update_sensors_fov_unittest.cpp](#) File Reference

Tests

- test group [f360_update_sensors_fov](#) ()
Tests to check the updated sensor properties with moving host and the azimuth properties within and beyond fov.
- [f360_update_sensors_fov_update_sensors_fov](#) ()
- [f360_update_sensors_fov_update_sensors_fov_azimuth_beyond_fov](#) ()

3.25.1 Detailed Description

Unit test to check if sensors_motion is updated in the sensor_properties

3.26 [source/unittest/rspp_update_sensors_time_unittest.cpp](#) File Reference

Tests

- test group [f360_update_sensors_time](#) ()
Test to see if the time_since_measurement is updated.
- [f360_update_sensors_time_update_time_since_measurement_in_sensor_props](#) ()

3.26.1 Detailed Description

Unit test to check time_since_measurement for each sensor is updated correctly.

3.27 source/unittest/rspp_update_sensors_unittest.cpp File Reference

Tests

- test group [f360_update_sensors](#) ()
Add brief description of test group, i.e. describe what functionality is tested. When using multiple test groups, make sure to write a brief description for each test group. The description should be unique and describe the specific scenario that is tested in that group.
- [f360_update_sensors_FunctionToTest_Descriptive_Tag](#) ()

3.27.1 Detailed Description

This file contains unit tests for content of f360_update_sensors.cpp file

3.28 source/unittest/rspp_update_sensors_valid_info_unittest.cpp File Reference

Tests

- test group [f360_update_sensors_valid_info](#) ()
A group of tests that check if valid sensors are updated in the tracker_info struct.
- [f360_update_sensors_valid_info_check_esr_valid](#) ()
- [f360_update_sensors_valid_info_check_srr_valid](#) ()
- [f360_update_sensors_valid_info_check_srr_esr_valid](#) ()
- [f360_update_sensors_valid_info_check_invalid_sensor](#) ()

3.28.1 Detailed Description

unit test to check if the f360_update_sensors_valid_info is updating the valid sensors in tracker_info

3.29 source/unittest/rspp_vcs_long_sorted_dets_support_functions_unittest.cpp File Reference

Tests

- test group [f360_clear_dets_vcs_long_sorted_info](#) ()
Tests to ensure that function Clear_Dets_Vcs_Long_Sorted_Info works as expected.
- [f360_clear_dets_vcs_long_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#) ()
- test group [f360_update_dets_vcs_long_ref_sorted_info](#) ()
Tests for Update_Dets_Vcs_Long_Ref_Sorted_Info(). Purpose is to test function Update_Dets_Vcs_Long_Ref_Sorted_Info() in all possible branches and extreme situations when detection is below/above all reference points in vcslong_sorted_ref_det_idx to an arbitrary valid index as this is expected to be untouched.

- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero](#)
()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_Zero_Prev_Re](#)
()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Over_All_Ref_Points](#)
()
- [f360_update_dets_vcs_long_ref_sorted_info_Test_Update_Dets_Vcs_Long_Ref_Sorted_Info_Valid_Det_Below_All_Ref_Points](#)
()
- test group [f360_sort_detections_vcs_long](#) ()
 Tests for Sort_Detections_Vcs_Long.
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_General](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_One_Detection](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_No_Detection](#) ()
- [f360_sort_detections_vcs_long_Test_Sort_Detections_Vcs_Long_Detection_Above_Largest_Ref_Point](#) ()
- test group [f360_get_first_relevant_long_sorted_det_idx](#) ()
 Tests for Get_First_Relevant_Long_Sorted_Det_Idx.
- [f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_General_Case](#)
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- [f360_get_first_relevant_long_sorted_det_idx_Test_Get_First_Relevant_Long_Sorted_Det_Idx_No_Dets_Above_Desired_Valu](#)
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3.29.1 Detailed Description

This file contains tests to verify that functions related to detections sorted in VCS-longitudinal order. It contains tests for both the actual sorting function and also support functions for efficient use of the sorted list.

[illegible]

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